AGE-BASED VARIATION IN EMOJI PRAGMATICS: INTERPRETATION AND INTERGENERATIONAL MISCOMMUNICATION

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BY

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CERTIFICATE

This is to certify that the thesis entitled **Age-Based Variation in Emoji Pragmatics: Interpretation and Intergenerational Miscommunication** submitted by **VAISHALI KONDUR (21IHML03)** in partial fulfillment of the requirements for the award of the degree of **Integrated Master of Arts (I.M.A.) in Language Sciences** from the University of Hyderabad is a bonafide research work carried out by her under our supervision and guidance.

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DECLARATION

I, **Vaishali Kondur**, hereby declare that this thesis entitled **Age-Based Variation in Emoji Pragmatics: Interpretation and Intergenerational Miscommunication** submitted to the University of Hyderabad in partial fulfillment of the requirements for the award of the degree of **Integrated Master of Arts (I.M.A.) in Language Sciences** embodies the veritable research work carried out by me under the guidance and supervision of **Prof. S. Arulmozi** at Centre for Applied Linguistics and Translation Studies, School of Humanities, University of Hyderabad. It is a research work which is free from plagiarism.

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Abstract

This dissertation examines age-based differences in emoji interpretation across four cohorts (18–25, 25–35, 35–45, and 45+) drawn from a multilingual Indian sample (n=70). Using a mixed-methods online questionnaire that combines Likert-scale items, contextualised interpretation tasks, and open-ended responses, the study investigates how different generations interpret five target emoji that vary in their degree of pragmatic reanalysis: the skull (💀), the loudly crying face (😭), the sparkles (✨), the slightly smiling face (🙂), and the wilted flower (🥀).

The findings document a systematic stratification of emoji interpretation by age. Younger cohorts assign ironic and pragmatically reanalysed meanings to these emoji, while older cohorts retain literal, iconic readings. The 25–35 cohort emerges as a transitional bridge group, displaying peak interpretive uncertainty and the sharpest single-step decline in active usage of sarcasm-dependent emoji. Relationship context emerges as the strongest contextual cue across all cohorts, exceeding sender age and platform of use. The study reports a 51.4% rate of self-reported misunderstanding of emoji in the sample. It proposes a three-level theoretical framework that draws on usage-based learning, cohort linguistics, and face theory to account for the observed patterns.

The contribution is twofold. Empirically, the study extends generational emoji research into a non-Western, multilingual context that has been underrepresented in the existing literature. Theoretically, it identifies the bridge cohort as a productive site for future research on intergenerational digital communication. It refines the contextual hierarchy that governs emoji interpretation in mixed-age communicative settings.

Chapter 1
Introduction

1.1. Computer-Mediated Communication

Computer-Mediated Communication (CMC) represents a change in modern communicative practices. It involves any form of human communication through electronic devices (Herring, 2004). This shift has changed how people interact across many digital platforms, including email, IM (Instant Messaging), social media, and other collaborative digital spaces. According to Herring et al. (2013), the field of CMC research is well established, with foundational work by Bieswanger (2013). This includes identifying critical grapholinguistic features characteristic of CMC at the structural, functional, and stylistic levels.

Bieswanger (2013) documents characteristic features of CMC, which include acronyms (OMG, LOL), written abbreviations (4u, ur), non-standard spellings, and emphatic punctuation (really?????, omg!!!!!!). These features operate at the grapholinguistic level (which encompasses the visual or written markers) and the stylistic level (including syntactic reduction and colloquial or dialectal expressions). Such features are not uniformly spread across all online/digital contexts. They instead appear more in private, informal communication, such as WhatsApp chats between friends/family, while remaining rare in formal contexts intended for unknown audiences or business-related communications (Dürscheid & Meletis, 2019).

Dürscheid (2005) and other theorists have established essential terminological distinctions between medium (technical infrastructure), form of communication (interactive modality), and text genre (communicative purpose). While modern mobile devices can perform functions almost as well as personal computers, the boundaries between the two have become increasingly blurred. The same device (a phone) may serve as the epicentre of formal use, social media communication, and ever-changing instant messages- all of which have vastly different norms for their communication. More interestingly, speed matters: rapid information exchange strips away formality, so texting feels more like speech than writing (Dürscheid & Meletis, 2019).

Herring and Dainas (2017) introduce the idea of “graphicons”, visual elements that include emoji, ASCII signs, stickers, and GIFs, within CMC systems. Among these graphicons, emoji hold a special status as the only visual elements formally integrated into the Unicode Standard. Unlike pictures or videos, emoji sometimes even work as substitute text components, appearing on similar lines as words and sometimes even substituting for them. (e.g., “I’ll come by 🚗” replaces the need for the verbal expression “car”). This substitution raises a fundamental question. Are emoji creating a novel writing system? Emoji are now everywhere, especially in apps like WhatsApp, where their use has piqued scholarly attention to their linguistic and communicative properties.

1.2. Social Networking

Social media platforms are among the primary modern spaces where we can observe linguistic innovation and variation in real time across different demographic groups. Platforms like X (formerly Twitter), Facebook, Instagram, Snapchat, and TikTok have become a cornerstone of daily communication for billions worldwide, fundamentally reshaping how language is used across these spaces. Each platform displays distinct features, yet all choose to participate in the same act of communication through varied styles and methods. These varied styles result in users developing distinct ways of communicating across platforms (Azad, et al., 2023).

Different generational demographics use language differently on social networking platforms. Researching platforms like X (formerly Twitter), Facebook, and Instagram across four generations (Baby Boomers, Generation X, Millennials, and Generation Z) has been studied to reveal that there are significant statistical differences in word frequency, levels of formality, and usage of emoji (Azad, et al., 2023). Boomers and Gen X show a stronger preference for formal patterns of language use. In contrast, Millennials exhibit greater informality and the use of abbreviations, and Gen Z is the demographic most comfortable with the integration of emoji and non-standard orthography. These generational differences reflect age and demographic-based differences in the communication styles the groups grew up with or are used to.

The phenomenon of using different communication styles based on platforms demonstrates the user's code-switching abilities. Users also change their linguistic and symbolic lexicons across platforms depending on factors such as the purpose of communication, the platform's rules or norms, and their audience. Facebook allows longer posts, so people are wont to write more detailed messages, whereas the character limit in a space like X limits people to shorter, more condensed messages. Instagram, on the other hand, is centred around photos and videos, where users communicate through them, as well as captions, emoji, and other context-specific communication strategies rather than fixed lexicons (Koch et al., 2022).

Research shows that the functions of social media are not limited to information exchange or communication, but also encompass identity construction, community formation, and the creation of emotional connections. Emoji play an important role in these social interactions and spaces, as users often use them to express group identity and build relationships (Kelly & Watts, 2015). Studies also show that the technologies people grew up with influence their long-term communication habits. These studies have important implications for organisations, and, as Azad, et al. (2023) emphasise understanding these habits and communication styles is necessary for designing spaces and messages that effectively engage different audiences.

1.3. What is an Emoji?

The term *emoji* derives from Japanese: 'e' (絵, picture) and 'moji' (文字, character) (Danesi, 2016). Emoji are a discrete category of graphic signifiers designed by Shigetaka Kurita in 1995 for a Japanese mobile carrier, later standardised by the Unicode Consortium[1] in 2010 and used globally across platforms. While emoji are frequently confused with their ancestral emoticons, they constitute a distinct semiotic category with substantial differences (Kelly, 2015). Emoticons are written representations that usually consist of punctuation marks and letters or numbers. In contrast, emoji are standardised visual icons that appear similarly across digital platforms, encompassing not only facial expressions but also objects, activities, gestures, and even abstract concepts.

[1]https://www.unicode.org/consortium/consort.html

Emoji achieved tremendous global success following their integration into iOS in 2011 and, subsequently, into Android. The Unicode Consortium's streamlining of emoji into standardised characters enabled consistency across platforms. More contemporary research shows that emoji usage has become the norm across all platforms, especially among younger users, who use them to express emotional nuances, tone, and intent that verbal language alone cannot convey. The universality of emoji has taken them from stylistic decorations to integral multimodal communicative components, warranting linguistic and semiotic investigation.

As emoji grew in number, people began asking uncomfortable questions, because a visual representation of a face, expression, food, or person was never just that. Why did we have six different sushi emoji but none for tacos or enchiladas? Why were all the professional emoji shown as men? Why were all the people in the emoji white? These questions were not about picture icons; they were about whose existence was worth representing digitally. By 2014, something known as the "Great Emoji Politization" had begun. Being asked, there were more and more questions regarding more serious ideas. Why is there an emoji for the Israeli flag but not the Palestinian flag? Why were emoji not showing same-sex parents or even single parents? Where were emoji for women with jobs? Or even a show for People of Colour? Emoji shifted from being fun additions to the digital space to becoming a new language of the digital age, but a language with blind spots nonetheless. Responding to growing pressure and advocacy, the Unicode Consortium began implementing more inclusive standards. In their revisions from 2017 onwards, we see the addition of every flag, women with babies, women wearing hijabs, representations of People with Disabilities, and gender-neutral figures to the emoji list. Moreover, this representation continued to grow over time. (WIRED, 2018[2])

From a semiotic perspective, emoji can function as iconic, indexical, and symbolic signs simultaneously within the Peircean framework (Peirce, 1931-1958). In terms of being iconic, a 🌻 (sunflower emoji) will simply represent a sunflower. In indexical value, emoji indicate emotions or reactions, like the 💀 (skull emoji) being used to express extreme amusement. Finally, as symbolic signs, emoji maintain their meanings through shared social conventions, such as the 👍🏽 (thumbs-up emoji) representing approval. This triple-dimensional nature of emoji allows them to form a complex communicative system rather than simple decorative or ornamental systems (Schneebeli, 2025; Halté, 2019).

1.4. Current Work Done on Emoji

Academic research on emoji has become prominent since approximately 2016, with contributions from many fields, including linguistics, psychology, cognitive science, computer science, and communication studies. However, although the number of studies has increased significantly, the quality and rigour of linguistic analysis remain inconsistent. Ferrari (2023) argues that though many studies treat emoji as legitimate linguistic units, they do so without sufficient empirical evidence. As a result, some research relies on assumptions from early studies, leaving important theoretical questions about emoji and language unanswered.

[2] https://www.wired.com/story/guide-emojifully/

Ferrari (2023) documents that, though most modern research on emoji shares a common understanding of the foundational concepts of emoji's status as a linguistic component, it fails to develop the in-depth linguistic analysis needed to ground emoji's linguistic properties fully. Key questions persist: Do they possess grammatical structure or just semantic content? Are there rules to govern how multiple emoji are arranged within a single message? Moreover, how do emoji interact with the structure of accompanying text? While studies on emoji pragmatics have identified many communicative functions: expressing emotion, modifying tone, indicating humour or irony, displaying attitudes, and acting as quick response signals, being among the functions, the syntactic structure of the emoji remains studied in limited amounts (Cohn, et al., 2019).

Robertson, et al. (2021) conducted a longitudinal study examining how emoji meanings changed over time from 2012 to 2018 using computational methods. Their findings showed that emoji meanings are not fixed or universal. Instead, emoji meanings shift over time, differ across platforms, and depend on context. For example, the "tears of joy" 😂 emoji, originally used to express laughter, is now used for more complex emotions, such as irony. The study also identified patterns in emoji meaning change, including "seasonal usage" (like the increased use of the basketball 🏀emoji during sports seasons), trends driven by internet memes which were very temporary (such as the frog emoji 🐸 used usually with the cup of tea emoji☕to show the famous gossip meme by Pepe the Frog), and major semantic shifts like the skull emoji 💀 which showed clear evolution from representing death to Halloween to signaling humour.

Research on generational differences in emoji interpretation (Zhukova & Herring, 2022; Kostadinovska-Stojchevska & Shalevska, 2024) shows that different age groups interpret emoji differently. Studies suggest that Gen Z users interpret emoji more creatively and ironically, diverging from designated Unicode definitions, whereas Millennials tend to interpret them relatively more conventionally or literally. Generations older than that, like Gen X or Baby Boomers, are sometimes likely to misunderstand them entirely. Schneebeli (2025) extends this line of work with a focused mixed-methods study of the skull emoji (💀), which is taken up in detail in Section 1.8.

1.5. Semiotics of Emoji: Foundational Frameworks for Emoji Analysis

Peirce's theory of signs (Peirce, 1931-1958) explains meaning as a relationship between three elements: the sign itself (the symbol you see, for example the skull emoji💀), the object (what the sign refers to, here, humour), and the interpretant (the meaning formed in the mind of the interpreter, which could range from death to humour). In this model, a sign represents something to someone in a particular context, and meaning emerges through individual interpretation (Merrell, 2005). Peirce also classified signs into three types relevant to understanding emoji: icons, indices, and symbols. Icons resemble what they represent, indices point to a particular state or reaction, and symbols rely on shared social understanding for meaning.

According to Peirce (1931-1958), signs can function as icons, indices, or symbols. Icons represent meaning through visual similarity to what they represent. For example, a flower emoji (🌼) simply represents a flower. Indices communicate meaning through context, often indicating emotions, attitudes, or references within a conversation. For instance, a pointing finger emoji

(👉🏾) may indicate direction or refer to something mentioned in the text. Symbols, on the other hand, rely on socially agreed-upon meanings rather than visual resemblance or direct connection to what they represent. An example is the sparkle emoji (✨), which is sometimes used to signal ironic emphasis, even though its visual form does not inherently suggest that meaning.

Herring and Dainas (2017) expand on semiotic analysis by examining how emoji function at the graphemic level within writing systems, arguing that emoji should be understood as part of the text itself rather than as a decorative element. This perspective establishes emoji as an element of modern written language, as a written component, warranting analysis of their syntactic, semantic, and pragmatic properties with rigour.

1.6. Semantics: Meaning-Making and Interpretative Processes

Semantic analysis of emoji is a foundational yet contested domain within emoji research. Semantics looks at relationships between symbols and their meanings. This includes denotative meaning, which is officially assigned, and connotative meanings, which develop through actual social use in naturalistic contexts. In the case of emoji, this is especially noticeable: the Unicode Consortium assigns officially designated tags to each emoji character (e.g., "skull emoji", "face with tears of joy"), but actual user interpretations diverge substantially from meanings sanctioned by the Unicode Consortium.

Emoji exhibit extensive semantic ambiguity, which Danesi (2017) characterises as a fundamental property of visual language. A single emoji can have multiple layers of meaning simultaneously. It may represent an object, evoke emotions or cultural associations, signal an attitude or particular reaction, and also be a part of meanings shaped by memes or internet culture. Halté (2019) argues that emoji never actually lose their iconic nature but can simultaneously function as indexical and symbolic signs.

The way emoji change meaning is similar to how language itself has evolved historically. Robertson, et al. (2021) showed this using corpus linguistics and word embedding analysis. Take, for instance, the frog emoji. It migrated beyond meaning "literal frog" to becoming a meme reference for the "but that's not my business" meme, then entered political spaces, and has since cycled through multiple meanings. The skull emoji's semantic transformation demonstrates an "S-shaped" change, a curve from literal to figurative usage within approximately 12 months (2013), suggesting that its meaning was reconstructed by rapid collective change (Robertson, et al., 2021; Schneebeli, 2025)

Riordan (2017) and Sampierto (2024) stress that the emoji's communicative role extends beyond expressing emotions into the realm of disambiguation. Emoji resolve the inherent semantic ambiguities of text-based communication, which lack paralinguistic cues. Emojis even serve as disambiguating resources that clarify a speaker's intent, emotional stance, and irony. This semantic role interacts with the grammatical and pragmatic aspects of communication. Danesi (2017) explains this by distinguishing between immediate and dynamic objects. Immediate objects are about the direct denotations of the emoji, while dynamic objects refer to the broader connotations it evokes. For example, the skull emoji directly represents "death", while also evoking broader social connotations of "laughing to death".

Ljubešic & Fišer (2016) show that the conventional meanings that are initially associated with emoji can influence how people process and comprehend these messages cognitively. To fully understand emoji semantics, it is important to consider both denotational meanings (basic or literal definitions) and connotational associations that develop through actual usage. In addition, emoji interpretation usually depends on context and pragmatic-dependent factors within communication.

1.7. Perception: Cognitive and Generational Variation in Emoji Interpretation

The way people perceive emoji is a complex cognitive process influenced by many variables, including the user's age, cultural background, platform experience, linguistic competence, and immediate communicative context (Zhukova & Herring, 2022). Initial research into emoji suggested that they might constitute a "universal language" (Danesi, 2017) that transcends cultural and linguistic boundaries. Still, research indicates substantial variation in how emoji are interpreted across demographic groups. Fundamental questions remain unanswered: do specific emoji raise the same interpretations consistently across diverse populations? How much does context reduce emoji ambiguity? How do platform-specific usages and renderings affect interpretation?

Research consistently shows that different generations interpret emoji differently, and this can affect intergenerational communication. Zhukova and Herring (2022) found that Gen Z members tend to interpret emoji more creatively and ironically, often assigning layered or nonliteral meanings. Contrastingly, Millennials generally interpret emoji more literally or according to conventions. Older generations are more likely to misinterpret the meanings of these emoji until they are explained. This difference between generations is often linked to each generation's socialisation with technology. Gen Z grew up with emoji already a solid part of their digital language repertoire, which makes them more intuitive and flexible in understanding and interpreting their meanings. Older users, however, were introduced to emoji later in life and so rely more on the official meanings assigned by the Unicode Consortium, resulting in more conservative interpretations. (Mannheim, 1952).

The interpretation of emoji is highly context-dependent. The skull emoji used in response to "My morning was crazy" may signal amusement, disbelief, or humour, while the same emoji used with a sentence like "I just heard about the disaster" may express shock or distress. This context-sensitivity is quite characteristic of how language works more broadly, where meaning arises from the interaction among the message accompanying the emoji, the context of its use, and the shared understanding of the people communicating. However, miscommunication can occur when people interpret the message's context differently. This risk is especially high in cross-generational communication, where people might have different interpretations and frameworks for emoji meanings (Udoudom, et al., 2024).

Platform-specific differences in emoji interpretation add another layer of complexity. The same emoji may look different across operating systems, browsers, and applications. An emoji that may look friendly on iOS might look more ambiguous or even negative on Android. People using different platforms may interpret the emoji differently because the visual cues they receive from each version differ. Although this technical aspect of the emoji is not very studied, it can influence how accurately or successfully emoji can communicate meaning (Herring & Dainas, 2017). Overall, understanding emoji perception requires consideration of many factors, including cognitive processes, cultural background, generational differences, and the technological systems through which emoji are displayed.

1.8. Move from Literal to Exaggerated Usage

One of the most significant processes observed in contemporary emoji use involves a shift from literal to figurative and exaggerated meanings, with greater prevalence among younger user populations. This semantic transformation is in line with historical linguistic processes documented in studies of semantic change and etymology, where expressions originally denoting concrete phenomena acquire metaphorical, ironic, or intensified meanings through repeated use in language cultures.

The skull emoji (💀) is the paradigm case. Originally designated for literal death or danger (skull imagery), contemporary usage predominantly signals extreme amusement (dying from laughter, laugh till I die), or even embarrassment or emotional overwhelm. This shift occurred very quickly, within approximately one year (2013), when the emoji's semantic associations shifted from a literal death reference to figurative amusement signalling (Robertson, et al., 2021). So far, this change appears substantially irreversible, with contemporary users predominantly using it for amusement signalling, whereas the literal meaning of "death" is comparatively rather rare (Schneebeli, 2025).

These conceptual mechanisms that drive literal-to-figurative semantic shifts in emoji mirror those documented in the linguistic metaphor theory. Users capitalise on the visual properties and the symbolic associations of emoji to construct novel meanings through extension and analogical reasoning. The skull emoji's transformation involves recognising that "dying" (metaphorically applied to extreme laughter "to the point of death") can be visually represented by imagery of death, producing an apt visual metaphor. On the flip side, analogously, the frog emoji (🐸) became associated with the meme "But That's None of My Business", which was originally a meme where the phrase functions as a sarcastic remark, along with the picture of Kermit the Frog from The Muppets, used to underscore criticism or mock socially questionable behaviour, often framed as detached or ironic commentary. (Kermit Sipping Tea/That's None of My Business, Know Your Meme, n.d.). Users eventually began using the frog emoji (🐸) in contexts involving judgment or commentary, allowing the meme to expand beyond the original, and the emoji's expansion into broader figurative application, which signifies users' understanding/knowing observation or subtle criticism (Robertson, et al., 2021)

Exaggeration and hyperbole constitute a particularly prominent figurative mode of emoji use among younger users, in which semantic intensity is amplified beyond what a literal interpretation would suggest. Multiple repeated emoji (💀💀💀💀💀) intensify emotional valence far beyond what a single emoji conveys. This creates gradations of amusement or distress through reduplication — a mechanism quite common across languages for signalling emphasis or magnitude (Cohn, et al., 2019). Repeated smiling faces convey heightened cheer; repeated food items convey the magnitude of an amount being described. Combinations of skulls (💀), tears of joy (😂), and other facial expression emoji create expressions that written language cannot achieve as readily (Cohn, et al., 2019). This compensatory function, emoji supplementing what written linguistic resources cannot easily express, illuminates their essential role in multimodal communication that lacks paralinguistic channels (Novak, et al., 2015).

Schneebeli (2025) documents that the functional shift of the skull emoji reflects ongoing pragmaticalization, in which lexical elements evolve into markers that convey the speaker's attitude towards the hearers or the message rather than its truth conditions (Aijmer, 1997). The skull emoji (💀) functions increasingly as a stance marker, moving away from the iconic function toward pragmatic roles. Research on TikTok comments also demonstrates that emoji that function as humour/awkwardness markers are replacing traditional punctuations (Kröll, 2022). The generation-specific characteristic of "literal-versus-figurative" interpretative differences carries particular significance: Generation Z (Gen Z) demonstrates that they can freely use emoji in any way- layered with ironic interpretations where emoji can simultaneously maintain both literal and semantic association while supporting figurative meanings. For example, the thumbs-up emoji (👍🏽), which conventionally signifies approval, is interpreted as passive-aggressive dismissal in Gen Z contexts. At the same time, they tend to retain the capacity to understand it as a literal positive acknowledgement, depending on its context. This flexibility of interpretation indicates some amount of cognitive sophistication in managing semantic ambiguity and marks emoji usage as a locus of linguistic variation and in-group identity construction.

Chapter 2
Review of Literature

2.1. Social Networking and Linguistics of the Internet

The rise of the internet as the primary medium of human interaction has fundamentally restructured the nature of communication. As Yao and Ling (2020) note in their introduction to a special issue of the Journal of Computer-Mediated Communication, the core question of the field has shifted from asking "what is computer-mediated communication?" to "what isn't?". This reflects how fully digital media has penetrated everyday social life. In the 25-or-so years since the journal's founding, computers have given way to smartphones, wearable technology, smart home devices, and autonomous systems, each restructuring how we communicate and what it means to communicate. As of 2019, 4.33 billion people actively use the internet, and American adults spend over 11 hours a day communicating on digitised platforms and media. According to a recent update, there are approximately 1 billion internet users in India, and the average time spent online per day is around 6 hours and 49 minutes. This again signals the extent to which digital communication has become central to how society operates and is also a dominant force in society. This unprecedented scale of use and pervasiveness in people's daily lives has prompted linguistics to establish a dedicated field of inquiry: Internet Linguistics.

Internet Linguistics, as defined by Crystal (2011), represents the scientific study of all manifestations of language within the electronic medium. This definition deliberately narrows our focus from broader terms such as Computer-Mediated Communication (CMC) or Digitally-Mediated Communication (DMC) to examine the linguistic phenomena in this medium rather than communication and all its forms. The field emerged from the recognition that the internet constitutes the largest language database ever documented, requiring linguists to apply established methods of description and analysis to an unprecedented corpus. Instead of treating "the internet" as a monolith, Crystal's framework acknowledges the stylistic and functional diversity of internet outputs, including email, chat, blogging, social media, and text messaging, each with distinct communicative properties, constraints, and linguistic expectations. Internet linguistics operates with the full functional capacity and apparatus of traditional linguistic inquiry, from syntax to sociolinguistics. However, the field still demands a reexamination of previously known facts about language. The need for reexamination stems from the fact that the properties of the electronic medium have fundamentally altered how language functions across various dimensions, such as the spoken, written, and graphological. The field of Internet linguistics is studied; it can be understood as an extension of linguistic science itself, using systematic methods to examine language in digital environments where features such as real-time interaction, massive data availability, rapid technological change, and questions of anonymity and access introduce both challenges and unprecedented research possibilities.

This saturation has generated parallel changes in linguistic practice. Scholars discussed in Yao and Ling's (2020) overview have noted that CMC researchers must shift their focus from digital spaces as objects to the enduring nature of human communicative processes mediated by these technologies. Instead of being distracted by the trends of new devices and platforms, researchers must ask what fundamentally changes, and what remains constant, in human communication when mediated through these new digital channels. This perspective is central to understanding emoji as a communicative phenomenon, since their use is not just a technological novelty but a response to persistent human needs: expressing emotion, modulating tone, signalling intent, and managing interpersonal relationships. All these functions that emoji fulfil cannot be fully expressed in digital discourse, unlike face-to-face natural human communication.

The intersection of social networking and language is most vividly illustrated by platforms such as WhatsApp, Instagram, Facebook, and Twitter. Arafah and Hasyim (2019), writing from a Saussurean linguistic perspective, argue that the rapid increase in internet access (in Indonesia alone, over 62% of internet users regularly engage with social media) has produced new forms of language that blend verbal and non-verbal elements. Their study, drawing on WhatsApp communications, finds that emoji form a system with syntactic, semantic, and pragmatic dimensions, functioning not just as decorative additions to texts but as linguistic elements integral to meaning-making. This argument, that emoji are part of language, not ornaments to it, sets the stage for any scientific inquiry into the semiotic properties of emoji and their communicative functions. Together, these foundational works establish a critical framing and understanding for research on emoji: any study of how people interpret emoji must account for the broader digital-linguistic context in which they are used, which includes the platforms on which they circulate, the socially accepted norms that govern their use, and the social relationships within which they acquire and even change meaning.

2.2 Emoji as a Semantic and Semiotic System

Whether emoji constitute a language or something language-like has been at the centre of scholarly debate since their widespread adoption. Applying Roland Barthes's distinction between langue (the abstract social institution of language) and parole (its individual, situated enactment), Arafah and Hasyim (2019) argue that emoji operate on both levels simultaneously. Emoji operate as a system- with their shared conventions, recognisable pictographic forms, and socially negotiated and understood meanings constitute a langue. The choices individuals make in using emoji in their specific conversations and contexts constitute parole. Because of this dual structure, emoji are neither idiosyncratic nor fully standardised: they are a social semiotic system that can be learned, negotiated, and innovated.

This frame of semiotics is extended by Lebduska (2014), who reads emoji through the lens of visual culture and rhetoric. Lebduska places emoji in the long tradition of attempts to supplement written language with visual cues, from the nineteenth-century proposal by writer Alcanter de Brahm for a *point d'irone* to the emoticons of early internet culture. Emoji, as argued by Lebduska, carry rhetorical power because they are not fully specified in their meaning. They invite interpretation, foster ambiguity, and mirror the communicative tics of their users. They symbolise an attempt to close the gap between intended and received communication. However, ambition is both the emoji's strength and its limitation: even the most recognisable pictograph is filtered through the reader's lens.

The trajectory of the skull emoji (💀) is a conceivably striking illustration of the emoji's semiotic complexity. Kostadinovska-Stojchevska and Shalevska's (2024) qualitative study of internet slang used by Gen-Z demonstrates how this single emoji underwent a radical pragmatic reanalysis. The skull emoji, originally associated with death, has been entirely repurposed by Gen Z users to express laughter, embarrassment, and hyperbolic disbelief, meanings that are drastically detached from their original visual references. The authors show, in their corpus of social media data, that the skull emoji (💀) can now function as a tone tag to signal that the preceding content is, in fact, meant to be an expression of humour, or even as a form of punctuation that adds structure to the flow of digital text. This dual capacity for signalling stance and organising discourse shows that emoji meanings are not only non-standard but also differ by user.

All these studies argue that the emoji cannot be simply reduced to an iconic system of universally understood pictographs. They are semiotic resources constructed through social negotiation, situated historically, and subject to constant reinterpretation. Any study of emoji interpretation must therefore contend with their inherent polysemy and community-specific conventions.

2.3. Semantic and Pragmatic Study of Emoji

Studying the semantics of emoji allows us to look at what an emoji means "on its own", while studying its pragmatics gives us a look into how its meaning depends on the context it is used in, including the intentions of the sender, the interpretation of the receiver, and its social acceptability within the community. Both strands of research have flourished in recent years, producing nuanced accounts of how emoji function beyond their face value designations.

The most rigorous empirical study of emoji semantics to date is Robertson et al.'s (2021) "Semantic Journeys: Quantifying Change in Emoji Meaning from 2012-2018". Using computational linguistics techniques and applying them to six years of Twitter (now X) data, we demonstrated what we now know as instinctive: emoji meanings are not static; they change systematically over time. They understand this as a process similar to semantic change in natural language. The researchers identified five patterns in the semantic development of emoji. They found evidence that fewer abstract emojis (those with a clear pictographic referent) are more susceptible to meaning change than their more abstract counterparts. External factors, such as seasonality and world events, are also documented in the context of specific emoji. This longitudinal study challenges the assumption that emoji meanings are fixed, transparent, and universally shared, a view that has been popular in academic discourse.

Kröll (2022) explores the pragmatic dimension of emoji use in depth in her study of laughter-based emoji in TikTok comment sections. Drawing on Conversation Analysis and multimodal interaction analysis, Kröll examines how emoji related to laughter are used to associate with, distance from, or modulate the register of humour content to which they respond. The study shows that in CMC, laughter is a complex social action: it can signal affiliation, diffuse awkward interactions, express uncertainty, or indicate detachment. Kröll argues that emoji encoding laughter inherits this pragmatic complexity while adding new layers of meaning enabled by their visuals. In CMC spaces where tone of voice and facial expressions are absent, emoji must do the expressive work that is the signature of face-to-face interaction.

Arafah and Hasyim (2019) find that emoji serve many communicative roles: they act as syntactic elements that sometimes occupy structural positions in messages, and as semantic units that contribute to propositional content. They also serve as signals of interpersonal relationships and communicators' stance. For example, birthday wishes often end with emoji that serve this interpersonal function, reinforcing solidarity and signalling the relational context of the exchange. These findings align with broader work in pragmatics that suggests language use is always simultaneously referential and relational.

In 2025, Schneebeli extends this pragmatic approach to the skull emoji (💀) explicitly. The author demonstrates how the emoji's function shifted from iconic (representing the death of Halloween) to indexical (expressing stance) through corpus analysis of English web data from 2018 and 2021. She finds an almost complete reversal between the two time periods: in 2018, iconic use dominated (55.6%), and by 2021, indexical use prevailed (56.6%). The skull emoji (💀) is now used to express critical reactions ironically or sarcastically. This microscopic analysis shows how emoji semantics and pragmatics are intertwined: the skull's new meanings are not arbitrary. They emerge from their previous associations with death and the macabre.

Lebduska (2014) argues that emoji occupy a "both/and" position, offering a rhetorically informed account of their pragmatics: they are used to clarify and mystify simultaneously, and to close and open interpretive gaps. Their success in a pragmatic context depends on shared context: between speakers who share history and cultural frames, an emoji can be a very efficient communicative tool, but between speakers who do not share certain dimensions of culture and history, they are a source of confusion and misunderstandings, a concept this paper returns to in later chapters.

2.4. Emoji in Computer-Mediated Communication (CMC)

Since the early 2010s, the study of emoji in Computer Mediated Communication (CMC) has grown rapidly. This generated a substantial body of empirical research on how, why, and when people use emoji, and how their use affects communication outcomes. Manganari (2021), in a critical review of forty-six empirical studies published between 1998 and 2020, provides an extensive synthesis of this work. Distinguishing between research from the users' perspectives (examining functions, frequency, demographic profiles, and conditions of use) and research from the perspective of the viewers (examining the effects of emoji on perceived emotion, message credibility, and interpersonal connection).

Manganari (2021) finds that emoji serve several well-documented functions in CMC: they signal playfulness, enhance social connectedness, convey positive emotion, and increase the perceived usefulness of messages. On the viewer's side, an emoji may enrich communication, convey positive attitudes, and hold attention. Conversely, she also notes a potential "dark side" of using emoji. For example, in formal or professional contexts, the use of emoji can lower the perceived credibility of a message and undermine an employee's competence in an assessment. This context dependence is crucial for the use of emoji. Ones that are appropriate and helpful in informal and personal contexts may not be so appropriate, or even counterproductive, or even be counterproductive.

Yao and Ling (2020) lay out a broader context for this work, arguing that CMC is a field where boundaries between mediated and unmediated communication are no longer significant in the way they used to be, and this is a consequence that must be grappled with in this field, due to the pervasiveness of digital media. Emoji cannot be studied in isolation from the technological ecosystems in which they operate. Simultaneously, emoji cannot be reduced to mere technological features; instead, they must be understood as resources communicators use to serve the enduring needs of communication.

In an article published in Cogent Social Sciences, Barinaga-López et al. (2022) take a broader view, examining the sociopolitical impacts of emoji in CMC from 2015 to 2021, and map the entire landscape of scientific knowledge production. They find that research on emoji is heavily concentrated in computer science (29%) and communication (15%), with social sciences and linguistics accounting for a much smaller share. They do this by drawing on a dataset spanning computer science, communication, social science, psychology, behavioural science, and linguistics. This cluster analysis reveals two major traditions of research: one concerned with the use of emoji in specific social, cultural, and other contexts, and the other with the automated mass analysis of emoji data for sentiment analysis and opinion mining. The communicatively oriented tradition, which is more relevant to the present study, includes work on emoji from across cultural and demographic groups, the dynamics of emoji-mediated interactions in interpersonal communication and the role of these emoji in identity construction.

2.5. Sentiment Analysis Studies on Emoji

A substantial strand of emoji research emerged at the intersection of computational linguistics, machine learning, and affective computing. Here, the goal is to use emoji as signals for automated sentiment detection in social media datasets. This strand of research examines emoji as indicators of emotional polarity (positive, negative, or neutral). Here, their distribution across millions of posts can be analysed to map public mood, evaluate brand sentiment, detect fake news, or monitor political discourse.

Barinaga-López et al. (2022) note that many of the articles in their review focus on sentiment analysis and automated opinion mining, which is the single most common research topic in the emoji literature. This type of research is primarily conducted in computer science, data analysis, machine learning, and marketing, and it generally relies on large corpora of data from social media platforms such as Instagram, Twitter (now X), and Reddit. This paradigm involves training ML models on emoji-labelled data to predict the sentiment of unlabeled posts. They usually treat emoji as ground-truth indicators of the emotional weight of the messages they accompany.

Robertson et al. (2021) intervene critically in this paradigm. Their longitudinal demonstration that emoji meanings drift over time has direct implications for sentiment analysis: a model trained on 2015 data may misclassify posts from 2021 if it does not account for semantic drift in emoji meanings. The same emoji may move from predominantly positive to negative contextual associations, or become specialised for specific emotions, over the lifespan of a deployed system. This work, therefore, constitutes both a contribution to computational semantics and an inherent critique of static approaches to emoji sentiment.

2.6. User Interpretation and Perception of Emoji

How people interpret emoji, and the factors that influence those interpretations, are among the most policy-relevant areas of emoji research. Earlier work assumed that emoji meanings are somewhat fixed and shared. But now, a growing body of evidence demonstrates that emoji interpretation is highly variable and shaped by many factors, such as personality, relationship context, cultural background, and your place within a generational cohort.

Völkel, et al. (2019) provide a particularly detailed account of individual-level variation in emoji interpretation in their study, “Understanding Emoji Interpretation through User Personality and Message Context.” They used the Big Five Inventory-2 (BFI-2)[3] to measure user personality and, with it, conducted an online survey with 646 participants, and asked them to add appropriate emoji to short text chats involving four types of receivers (parent, friend, colleague, romantic partner), and four situational variations (informational, arrangement, salutary, and romantic). The survey findings show that an individual’s personality influences the choice of emoji. For example, agreeable users were more likely to use the heart emoji (❤️) even in negative situations, and openness predicted a greater variety of emoji choices. Importantly, even when participants were shown emoji found to be semantically similar in prior research, they still generated highly varied and rich interpretations, showing that emoji meaning is actively constructed in context rather than decoded from a fixed system of signs.

The same authors also highlight the role of the recipient’s context in shaping the use and interpretation of emoji. The same sender adjusts their emoji choice depending on their relationship with the recipient. This finding resonates with classical sociolinguistic accounts of style-shifting and audience design. This finding implies that emoji interpretation is a social and relational practice in which the communicator's identity and the norms of the relationship with the recipient play central roles.

In their study of generational differences, Zahra and Ahmed (2025) extend this analysis to the dimension of an age cohort. They surveyed Gen Z, Millennials, and Baby Boomers. They found that Millennials tend to interpret emoji at face value, where a thumbs-up emoji (👍🏽) is read as approval and a smiley face (☺) as happiness. Conversely, Gen Z routinely use emoji ironically, using the skull emoji (💀) to express the hilarity of something and the upside-down smiling face (🙃) to signal sarcasm or passive-aggression. Finally, they seem to have found that Baby Boomers often interpret emoji based on their iconic meanings, and that they have difficulty recognising non-literal, ironic, or community-specific uses that have become conventional among younger generations of users. These differences are bound to have consequences where this cross-generational communication is likely to break down in various spaces like the home, the workplace, or in other unpredictable settings.

Lebduska (2014) almost prophesies these findings in her rhetorical analysis, where she argues that emoji interpretation is always context-dependent. Even the most seemingly transparent emoji can be misread when the interactive contexts of the sender and receiver diverge. She illustrates this with examples from life, with a sad face on a tax return, a smiley face next to an “EXIT” sign in an emergency, in which there is a mismatch between the emoji’s conventionally agreed-upon meaning and its actual deployment in context. This produces confusion, rather than clarity. The relational and contextual assumptions that make emoji sensible are also the source of their interpretive ambiguity.

2.7. Emoji Usage Patterns and Evolutions

Emoji usage has changed over time across platforms, populations, and cultural contexts. Research on these changes reveals a rapidly shifting communicative landscape. The emoji

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[3]https://www.colby.edu/academics/departments-and-programs/psychology/research-opportunities/personality-lab/the-bfi-2andandandand/

repertoire is once again confirmed as a rapidly changing, continuously expanding code, with individual emoji meanings in a constant state of change and development.

We return to Robertson et al.'s (2021) longitudinal study of Twitter data, given its status as the most rigorous quantitative account of emoji semantic evolution. It uses word vector models that were adapted for emoji to trace "semantic journeys" (LOL) of individual emoji over time and identify patterns of development, and have had these patterns split into five: stability, gradual drift, rapid shift, bifurcation (here, an emoji acquires multiple distinct semantic associations), and convergence (where uses that were previously distinct end up merging). The study found that the Santa Claus emoji (🎅), for example, remains stable semantically across all six years of the study, while the face with tears of joy or laughter emoji (😂) undergoes a gradual but consistent shift where it is increasingly used in contexts that are evaluative rather than solely emotional. These findings have important implications: that emoji-based sentiment classifiers that do not account for such drifts will systematically produce errors.

Schneebeli (2025) extends Robertson et al.'s longitudinal approach to a single emoji: the skull emoji (💀) by using the enTenTen18[4] and the enTenTen21[5] web corpora. The study finds that between 2018 and 2021 (when Robertson et al. stopped counting), the use of emoji increased dramatically (by 144% in the earlier corpus). This increase reflects greater use of emoji on social media platforms. More substantially, there is a documented functional shift. Iconic uses of the emoji (representing death, Halloween, the macabre) declined from 55.6% to 38.4%, while indexical uses (expressing stance, affect, and evaluation) increased from 41.9% to 56.6%. Within the indexical category, evaluative uses grew sharply, with negative evaluations increasing from 18 to 51 instances. The author interprets this as evidence that the skull emoji (💀) is one in transition. That means that the emoji is undergoing the kind of pragmatic reanalysis that Robertson et al. understand and document at the lexical level.

We return momentarily to Manganari (2021), who finds gender-based, generational, and frequency differences in emoji use between formal and informal contexts. Usage frequency and the size of the user's repertoire are positively correlated with digital literacy and familiarity with platforms. However, the author notes that the causal leaning direction of these relationships is tough to establish from existing cross-sectional studies.

Ilona Kröll (2022) conducted a qualitative analysis of emoji in laughable contexts in the comment sections of TikTok. This study reveals that users deploy emoji strategically to manage associations and what they do not associate with. The users are reported to use the skull emoji (💀) and emoji adjacent to it to signal they find something funny to the "point of dying". This same usage of the emoji also signals their part in an in-group and as a tool of social bonding. She argues that the comment section is a distinctive CMC context that shares qualities of both private messaging and public broadcasting. Given the dual nature of the comment section, users are expected to strike a balance between expressiveness and legibility, ensuring these comments are usable and understandable to other users.

[4] https://app.sketchengine.eu/#op

[5] https://app.sketchengine.eu/#open

In Zahra and Ahmed's (2025) study of the varied use of emoji across generations, such as Gen Z, Millennials, and Baby Boomers, they find that Gen Z users have the largest emoji repertoires. In contrast, Baby Boomers use a smaller, semantically conservative set of emoji. Millennials occupy an intermediate space here as well, using emoji primarily to express positive emotions and social warmth. These generational differences are not merely preferences, but also reflect the fundamentally different processes that each generation uses to deploy emoji. The authors claim that this difference is due to Gen Z having adopted emoji as a primary means of communication during adolescence. In contrast, Baby Boomers adopted them as supplements to pre-existing communicative habits formed in a non-digital world.

2.8. Limitations and Gaps in Existing Research

Despite the growing popularity of emoji and the ever-expanding body of research on the topic, a few important limitations and gaps emerge across the surveyed literature. These gaps are also relevant to the present study, which examines age-based differences in the interpretation of emoji across four generational cohorts.

First, the existing literature on generational differences in emoji interpretation is methodologically thin. Zahra and Ahmed (2025) and related work provide useful preliminary findings. Still, these studies tend to treat generational categories as homogenous, combining Gen Z, Millennials, and Boomers as unified groups without accounting for variations within each generation. These variations may stem from differences in digital literacy, platform use, and communicative context. The four-cohort framework employed in the present study (18–25, 25–35, 35–45, 45+) offers a more granular breakdown than most prior work.

Second, though many studies document the presence of emoji misunderstandings (Manganari, 2021; Völkel, et al., 2019; Zahra & Ahmed, 2025), there is relatively less systematic research on the mechanisms that produce these misunderstandings, and on the conditions or contexts under which they are repaired versus left uncorrected. Manganari (2021) calls for work that links emoji use to specific individual profiles, but few studies have operationalised this call in a way that clarifies the process of cross-group miscommunication.

Third, while some authors, such as Völkel, et al. (2019), document the importance of the recipient's context in shaping emoji use, their study focuses on the sender rather than the recipient and uses hypothetical scenarios rather than naturalistic communication data. Future research can examine more carefully how shared history, role expectations, and communicative norms within specific dyads and groups shape the interpretation of ambiguous emoji.

Finally, the body of work in emoji research is heavily skewed towards English-speaking and North American or European populations, with much less work on emoji use and interpretation in non-Western or non-English-speaking populations (Barinaga-López, et al., 2022). Although several studies claim a cross-cultural scope, most concentrate on Western individualist contexts. The present study is conducted in an Indian context with a multilingual, culturally diverse sample and is positioned to contribute to broader international comparative research.

Together, these limitations show the continued importance of empirical, context-sensitive, and demographically attentive research on emoji interpretation. This study seeks to address some of these gaps by examining age-based differences within a single social context, attending to relationship variables, and exploring the mechanisms of pragmatic reanalysis that may produce the systematic misunderstandings across generational lines.

2.9. Theoretical Framework

Drawing on the foundations established above, this study adopts a three-level theoretical model to account for age-based differences in emoji interpretation across the four cohorts. Each level addresses a distinct scale of analysis: individual meaning acquisition, cohort-level norm formation, and intergenerational communicative breakdown. Together, the three levels explain why generational differences in emoji interpretation may emerge systematically from the conditions under which different cohorts first encountered emoji as a communicative medium.

Level 1: Individual Pragmatics: Usage-Based Meaning Acquisition

At the individual level, the study draws on usage-based theories of language learning, in which speakers build interpretive frameworks through repeated contextual exposure rather than from formal definitions (Tomasello, 2003; Bybee, 2010). On this view, users do not read emoji meaning from a Unicode specification; they construct it from observed instances of use. This acquisition process produces systematically different outcomes depending on the user's formative digital environment, whether the dominant register encountered was ironic or literal.

Research demonstrates that even when the same emoji is presented in an identical message context, users assign rather different meanings to it, with interpretation varying based on the user's personality, platform experience, and prior exposure (Völkel, et al., 2019). Most notably, in the present study, older adults have been found to interpret emoji more literally, consistent with emotion-mapping (Herring & Dainas, 2020; Liu, et al., 2024) based on the emoji's visual form. Younger adults, on the other hand, apply ironic, non-literal understandings. This difference in usage is not a comprehension failure, but a result of different learning trajectories: the present study confirms that interpretation norms are contextually acquired rather than formally taught.

Level 2: Cohort Linguistics: Generational Norm Formation

At the cohort level, this study draws on Mannheim's (1952) theory of social generations, and specifically his concept of "fresh contact". Mannheim argues that younger people encounter established cultures from a fresh perspective that is not yet as deeply shaped by personal experience as it is for older people, allowing them to reorient their experience and understanding of culture in ways adults cannot. Applying this understanding of fresh contact to digital communication, it stands to reason that the 18-25 cohort that came of age during the 2012-2016 Tumblr and Twitter (now X) irony adoption era encountered emoji at the exact time they were starting to normalise the emoji's sarcastic use, and their fresh look into the internet at this time shaped their usage and interpretive frameworks solidified around these norms.

This is substantiated by survey evidence that confirms the cohort outcome: Gen Z uses emoji with the most ironic and layered meanings, Millennials occupy a transitional, middle position, and Baby Boomers interpret the emoji most literally (Zahra & Ahmed, 2025). The 18-25 cohort respondents in the present study's sample have been identified as operating through a pragmatic, sarcasm-first framework. This represents the cohort whose fresh contact occurred at peak irony adoption. Lenneberg's (1967) Critical Period Hypothesis offers a complementary neurological rationale: plasticity in terms of interpretation is at its highest in adolescence and early adulthood. This means that norms acquired during this window are resistant to later revisions.

Level 3: Intergenerational Miscommunication: Asymmetric Context Hierarchies

The final level in this theoretical model is cross-generational interaction, and this study employs Brown & Levinson's (1987) Face Theory and the concept of asymmetric context models to explain communicative failures. Face theory holds that communicative acts are evaluated against shared assumptions about positive face (the desire to be approved of) and negative face (the desire for autonomy). When cohorts hold incompatible implicit assumptions (e.g., whether an emoji is face-saving or face-threatening), the same symbol (here, an emoji) can simultaneously convey solidarity for one party and signal passive-aggression for another. This is documented in the generational split over the thumbs-up emoji (Zahra & Ahmed, 2025).

A precise empirical account of this dynamic comes from Cui (2022), who found that younger adults perceive both sender age and the sender-receiver relationship as significant cues when interpreting ambiguous emoji-based statements. In contrast, older adults view sender age as mostly irrelevant and rely more on the relationship between communicators. This asymmetry is the mechanism behind the present study's key finding: the relative weight that recipients give to relationship context versus sender age. Where these hierarchies of context differ across cohorts, miscommunication is the structural outcome rather than the exception.

One boundary condition must be acknowledged. Some studies do state that there is very little variation in the interpretation of facial expression emoji, suggesting that highly conventionalised, visually transparent emoji have achieved cross-generational stability. Even such claims, however, admit contextual variations might still exist. The divergences documented in this study concentrate on all kinds of emoji. The study has chosen two emoji that are understood to be conventionalised (the crying emoji (😭) and the slightly smiling face emoji (🙂)), two emoji with usage concentrated within a specific age group (the skull emoji (💀) and the sparkle emoji (✨) and one emoji with less established meaning to explore variation in interpretation across age groups.

Chapter 3
Methodology

3.1. Research Design and Approach

This study aims to examine generational variation in emoji interpretation with a more pointed approach than has typically been taken in the existing literature. Instead of unifying every generation under broad demographic labels, the study uses four specified age groups, yielding a more granular account. The study is a cross-sectional, mixed-methods investigation, designed to examine age-based differences in emoji interpretation across four cohorts: 18–25, 25–35, 35–45, and 45+. The mixed-methods approach is appropriate because neither qualitative nor quantitative analysis alone would account for the full picture. Quantitative analysis aims to identify patterns in sarcasm use, emoji frequency, and interpretation. The qualitative side, drawing on open-ended responses, produces accounts that resist the tendency of quantitative analysis to flatten nuance. A combination of both helps the study provide a more comprehensive description of the data.

The foundation of this study lies within a post-positivist paradigm. Post-positivism holds that, while a social world exists outside and independent of a researcher's perceptions, our access to it is always mediated by the environment, language, social context, and the interpretation of that world (Guba & Lincoln, 1994). Post-positivism is an appropriate paradigm for this paper on emoji interpretation, where the object of study is social and variable. The study does not assume emoji to have fixed, observer-independent meanings that can be directly measured. Emoji meaning is treated as socially shaped and context-dependent, and uses survey-based numerical measures as indirect ways to capture these meanings. Likert scales are used because they are a standard method for measuring abstract constructs such as attitudes and behaviours that cannot be directly observed.

A cross-sectional study design was chosen for practical and theoretical reasons. Practically, the resources and timeline for this study did not permit repeated-measures data collection over an extended period. Theoretically, the phenomenon of interest here (whether different age cohorts interpret emoji differently at a single point in time) is fundamentally synchronic. The study's primary claim is that cohorts socialised into emoji use at different points in their lives have each developed structurally distinct interpretive frameworks. A cross-sectional comparison of all these age cohorts is the most appropriate design for investigating this claim. It is, however, acknowledged that this cross-sectional design cannot distinguish between age (patterns attributable to maturation) and cohort (patterns attributable to general socialisation) effects. This limitation is further discussed in Section 5.4.

The quantitative strand of the design draws on standardised Likert scale survey items to measure emoji usage frequency, self-reported communicative functions (tone clarification, emotional intensification, sarcasm, face-softening), and perceptions of emoji meaning stability and context-dependence. These items produce data that can be meaningfully compared across all age groups using descriptive statistics and cross-tabulation. The qualitative strand of the study draws on open-ended questions, responses and contextualised interpretation tasks in which respondents were asked to interpret emoji embedded in sample sentences. These responses were subjected to manual semantic coding, which produced categories that map onto the three-level theoretical framework. The integration of the two strands occurs at the level of interpretation: quantitative patterns show where divergence exists across cohorts, and qualitative responses help explain the pragmatic mechanisms that drive it.

Five emoji have been selected to represent a range of semantic types. First, the skull emoji (💀) was selected as the paradigmatic case of pragmatic reanalysis (here, we hope to replicate the results of a previous study to anchor the present study) as an emoji that has undergone the most dramatic and well-documented semantic shift in existing literature (Robertson et al., 2021; Schneebeli, 2025). Second, the loudly crying face emoji (😭) and the slightly smiling face emoji (🙂) were selected as cases of semantic inversion, where the relationship between visual form and communicative meaning has been partially reversed in younger cohorts. Third, the sparkle emoji (✨) was selected as a potential generational marker, and finally, the wilted flower emoji (🥀) was selected for its youth-specific connotation, which was expected to produce maximal uncertainty among older respondents. Together, these emoji span a range from the highly conventionalised to the highly innovative, allowing the study to test the hypothesis that divergence concentrates in pragmatically reanalysed emoji rather than in visually transparent ones.

3.2. Research Objectives and Research Questions

This study focuses on two primary research objectives, each addressing a distinct dimension of age-based variation in emoji interpretation. The three-tier theoretical framework outlined in Section 2.9 informs both. Usage-based learning theory grounds the analysis at the individual level. Mannheim's cohort linguistics provides the account of generational norm formation. Brown and Levinson's Face Theory provide the interpersonal foundation for treating emoji pragmatic reanalysis as a politeness strategy for managing face-threatening acts through irony.

Research Objective 1

To examine age-based differences in the semantic interpretation of emoji, specifically the distinction between literal (iconic) and pragmatically reanalysed meanings.

With this objective, the study investigates whether age cohorts systematically differ in their inclination to assign iconic (literal) meaning rather than pragmatically reanalysed interpretations to the same emoji in identical linguistic contexts. The theoretical expectation, drawing on individual pragmatics, is that younger cohorts who adopted emoji during a period of rapid pragmatic innovation are more likely to deploy non-standard meanings, such as sarcasm markers or face-threatening acts softened through irony. Older cohorts, for whom emoji represent a relatively recent addition to their communicative repertoire, are expected to favour more semantically transparent, iconic readings. This difference would reflect the workings of cohort effects: distinct generational norms for emoji meaning-making that solidified during periods of rapid innovation, where these competing interpretations are used by many age groups simultaneously. It also engages Mannheim's idea of "fresh contact", which posits that younger people, having less exposure to pre-existing cultures, are more likely to develop interpretive frameworks at variance with those of older generations.

Research Objective 2 (Exploratory)

To explore whether divergence in emoji interpretation correlates more strongly with current age or with the developmental life stage at which emoji were first adopted.

This objective engages Mannheim's argument that people learn the specifics of language use at formative developmental stages, after which they remain relatively stable and resistant to change. It also engages the hypothesis that pragmatic reanalysis of emoji is most readily acquired during periods of high linguistic plasticity, such as late adolescence or early adulthood. If this is correct, individuals who adopted emoji during this window would be expected to retain their reanalysed interpretations and the associated face-management strategies throughout life, regardless of subsequent ageing. Though a broad recall item was included in the questionnaire, the responses were not sufficiently detailed or reliable to support formal adoption-timing analysis.

This objective is treated as exploratory rather than as a directly testable hypothesis. The survey instrument used in this study was not designed to capture detailed adoption-timing data — that is, the precise age at which respondents first began using emoji and the social context of that adoption. The data, therefore, cannot directly compare age-of-acquisition effects against current-age effects with the precision required for confirmatory testing. The data can examine whether interpretive patterns across cohorts are consistent with the predictions of an age-of-acquisition account, and whether the bridge cohort identified in the analysis displays features consistent with such a transition. Stronger tests of this objective would require either retrospective adoption-timing data or a longitudinal design. This is acknowledged here at the design stage rather than presented as a deficiency in the conclusion.

Hypothesis

From Research Objective 1, the following hypothesis was derived. Younger cohorts (18–25) will assign pragmatically reanalysed and non-literal interpretations to sarcasm-dependent emoji at higher rates than older cohorts (45+), reflecting the operation of cohort-specific norms for face management through indirection and irony. Older cohorts are expected to show a preference for literal, iconic interpretations, consistent with less extensive prior exposure to pragmatic reanalysis in their own social networks. This hypothesis is grounded in Mannheim's fresh-contact theory and the related claim that cohort-specific pragmatic norms develop during periods of rapid innovation and persist as generational markers.

This hypothesis is treated as a theoretically informed expectation rather than as null-hypothesis significance-testing propositions in the strict statistical sense. Given a sample size of n=70, the study lacks sufficient statistical power for reliable inferential testing at the individual-cohort level. Accordingly, the hypothesis is evaluated through descriptive statistics, cross-tabulations, and comparative analyses, with qualitative evidence used to assess interpretive plausibility and to explore mechanisms underlying the observed patterns. This approach is consistent with the study's mixed-methods epistemological framework and follows conventions established in comparable qualitative-led studies of emoji and computer-mediated communication pragmatics.

3.3. Study Variables and Operational Definitions

The study involves several key variables, each requiring careful definition and measurement to capture the concepts under study accurately. The following section explains each variable and the rationale for its measurement.

Age Cohort

Age cohort is the primary independent variable, defined as the self-reported age bracket respondents selected from four options: 18–25, 25–35, 35–45, and 45+. These brackets were chosen to align with the previously defined theoretical framework. The 18–25 group represents the cohort whose digital socialisation occurred during the 2012–2016 irony-adoption era on Tumblr and Twitter; the 25–35 group represents a transitional cohort with partial exposure to the ironic norms; the 35–45 group represents users who adopted emoji as adults with pre-existing literal communicative habits; and the 45+ group represents the oldest cohort, whose digital socialisation occurred latest and who are most likely to rely on iconic or indexical emoji meanings. These four categories were retained from the survey's original design rather than being coded post hoc to preserve the integrity of cohort comparisons.

Emoji Usage Frequency

Emoji usage frequency is defined as the respondents' self-reported frequency of using each of the five-target emoji, measured on a five-point Likert scale where 1= Never, 2= Rarely, 3= Sometimes, 4= Often, 5= Very Often. A derived variable, active usage, was computed by combining the Often and Very Often response categories, yielding a binary classification of each respondent as an active/non-active user of each emoji. This threshold was selected because it captures meaningful and habitual usage while excluding ambivalent or experimental usage that is captured by the "Sometimes" category.

Emoji Interpretation

Emoji Interpretation is the study's primary dependent variable. It was defined in two distinct ways. First, respondents were asked open-ended questions about what they personally meant when sending each target emoji. These responses were manually coded into semantic categories (e.g. humour/irony, literal death, passive aggression, genuine warmth, uncertainty). Second, respondents were presented with five sentences containing a target emoji as contextualization tasks, each pairing a target emoji with a naturalistic sentence, and asked to select the most appropriate interpretation from a fixed set of options that included "ironic", "literal", and "ambivalent" readings. These definitions together capture both the production-side meaning (sender intention) and the reception-side meaning (receiver interpretation), which are essential for examining the miscommunication dynamics described in level 3 of the theoretical framework.

Sarcasm Use

Sarcasm use was operationalised as respondents' self-reported agreement with the statement "I use emoji sarcastically," measured a second time on a five-point Likert scale. For analysis, respondents who scored 4-5 were classified as high-sarcasm users, and those who scored 1-2 were classified as low-sarcasm users. This variable was treated as both a dependent variable (to examine whether sarcasm use varies by cohort) and an explanatory variable (to examine whether sarcasm use predicts ironic interpretation patterns).

Contextual Interpretive Cues

Contextual interpretive cues include sender age, sender-receiver relationship, and platform of use. These were defined as binary yes/no responses to direct questions about whether each cue affects emoji interpretation, supplemented by open-ended follow-up prompts that asked respondents to explain how each cue affects their interpretation decisions. These responses were used to assess the relative weighting of different context variables across cohorts.

Face-saving and Emotional Functions

Additional Likert-scale items measured the extent to which respondents use emoji to soften criticism, intensify emotion, and clarify tone. Such items were included to capture the dimensions of face theory as assigned by the framework.

3.4. Tools and Instruments Used

The primary instrument for data collection was a structured online questionnaire comprising forty-four items across five thematic blocks. The questionnaire was designed for the study and built in Google Forms, selected for its accessibility across devices, ease of distribution via social platforms, and conditional question routing. The questionnaire was administered in English. While the sample is multilingual, with respondents reporting Telugu, Hindi, Tamil, Odia, and Marathi as primary languages alongside English, English itself is selected as the instrument language because it is the primary language of emoji-mediated social media communication among the target population, as confirmed by platform usage data, which shows English as the dominant social media language across all age groups in the sample.

The questionnaire was structured into five thematic blocks. Block 1 collected demographic information, including age bracket, gender, education level, and primary language. Block 2 collected data on general emoji usage patterns, including usage frequency, platforms used, and daily hours spent on social media. Block 3 presented the five-target emoji individually and collected usage frequency data, open-ended questions about intended meaning, and responses to the five-sentence contextualization tasks embedded within them. Block 4 asked nine Likert-scale perception statements addressing an emoji's communicative functions, stability of meaning, context dependence, and sarcastic usage. Block 5 collected data on sender age and relationship effects through direct yes/no questions and open-ended explanations.

The sentence contextualization tasks in Block 3 were designed to elicit naturalistic, ambiguous utterances that lead to real-world intergenerational miscommunication. The five target sentences were:

1. "That presentation was amazing 💀"
2. "Stop😭"
3. "Just finished my taxes ✨"
4. "Thanks for sending that report 🙂"

5. “He thinks he’s cool like that 🥀”

Each sentence was selected because its literal and ironic readings are coherent and plausible, creating genuine interpretive ambiguity. This design follows established precedent in emoji pragmatics research (Cui, 2022), in which contextualised stimuli are used to reveal how respondents comprehend emoji.

Data analysis was conducted using Python with the pandas library for quantitative tabulation and a manual qualitative coding scheme for open-ended responses. Cross-tabulations were computed using pd.crosstab()[6] and normalised by row to produce percentage distributions within each age group, enabling fair comparison across cohorts of unequal size. Active usage scores were computed as the sum of “Often” and “Very Often” responses. Open-ended responses were read in full, and recurring semantic themes were identified inductively before being organised into a coding frame. All coded responses were verified against the frame to ensure consistency, and edge cases were resolved by reference to contextual cues within the response.

3.5. Ethical Considerations

This study was conducted in accordance with standard ethical principles meant for social science research involving human participants. As a questionnaire-based investigation of communicative behaviour, a low-risk topic that does not involve sensitive personal information, vulnerable populations, or experimental manipulation, no formal ethics committee review was required under the institutional guidelines governing this dissertation. Nevertheless, several ethical considerations were addressed proactively throughout the research.

Informed consent was obtained from all participants through a cover page at the start of the questionnaire. This page briefly explains the questionnaire's contents, the study's purpose, the voluntary and anonymous nature of participation, the intended use of the data for academic research, and the absence of any personally identifiable information in the final report. Participants were informed that they could withdraw at any time without consequences and that submitting the completed questionnaire would constitute confirmation of their consent. The questionnaire did not collect names, e-mail addresses, phone numbers, or any other directly identifiable information.

Anonymity was maintained throughout the data collection and analysis. Responses were stored in a password-protected Google Drive account accessible only to the researcher. When open-ended responses are quoted in this dissertation, identifying details have been removed or generalised where necessary to protect the confidentiality of the respondents. No attempt was made to reidentify participants, and the data were used solely for the analytical purposes that are described in this study.

The study’s sampling strategy raises a fairness consideration that warrants acknowledgement. By recruiting participants through social media platforms and personal networks, the study implicitly favoured respondents who are already active social media users. This population is, by definition, younger, more educated, and more digitally literate than the general population. This

[6] pandas.crosstab() is a function in the Pandas library used to compute a cross-tabulation (or contingency table) of two or more categorical variables.

sampling bias is discussed in Section 5.4. The researcher deliberately sought to mitigate this bias by distributing the questionnaire through multiple channels and specifically soliciting responses from older age groups via family and professional networks. Despite these efforts, the 45+ cohort remains the smallest in the sample (n=9), and this limits the generalizability of findings specific to that group.

No deception was used at any stage of this research. Participants were not misled about the nature or purpose of the study, and no debriefing was required. The survey stimuli (five naturalistic sentences containing the study's target emoji) are of the type commonly encountered in everyday digital communication, and their use in the questionnaire context posed no risk of psychological harm or distress. The study data will be retained in accordance with institutional requirements for dissertation research. They will not be shared with third parties or used for any other purpose outside of this study.

3.6. Data Collection

3.6.1. Participant Selection and Sampling

This study employs a convenience sampling strategy that is also non-probability. It is also supplemented by purposive elements designed to ensure representation across the four target age cohorts. Convenience sampling was the primary mode of recruiting respondents: the questionnaire was distributed through the researcher's personal Instagram and WhatsApp accounts and shared on online forums like Reddit. This approach was selected due to its feasibility within the time constraints of a short-term dissertation project, its effectiveness for reaching digitally active populations, and its alignment with precedent in comparable emoji research studies operating at similar scales (Manganari, 2021; Völkel, et al., 2019; Zahra & Ahmed, 2025).

A purposive sampling approach was introduced to address the anticipated underrepresentation of older age cohorts in a convenience sample drawn primarily from social media. Acknowledging that the 35-45 and 45+ cohorts are less likely to respond to social media-distributed surveys, the researcher actively solicited responses from older participants through personal networks. This dual-mode distribution strategy for social media among younger users and personal solicitation among older participants reflects a quota-informed sampling approach that sought to achieve a minimum cell size of n=9 across all four cohorts. The final distribution of n=28 (18–25), n=17 (25–35), n=16 (35–45), and n=9 (45+) shows partial success: the youngest cohort is overrepresented relative to the others, and the oldest cohort remains small. These imbalances are addressed analytically through row normalisation (see Section 3.6.4) and are acknowledged as a limitation in Section 5.4.

The total sample size of n=70 is consistent with the scope of a student dissertation and is comparable to studies of similar design in the emoji and CMC literature. It is sufficient to support the descriptive and cross-tabulation analysis that constitute the study's primary analytical strategy, and large enough for meaningful qualitative analysis of open-ended responses. It does not, however, support inferential statistical testing at the level of individual cohort comparisons with adequate statistical power, and no such tests are reported. The findings of the study are therefore presented as pattern-level observations and theoretical elaborations instead of statistically generalizable claims.

The study targeted adult participants (18 years of age or older) who are active users of at least one social media platform. No formal exclusion criteria were applied beyond the age minimum, and participants were not screened for proficiency in English, owning smartphones, or prior knowledge of emoji. The decision not to screen for prior knowledge was deliberate: the study aims to capture naturalistic variation in emoji interpretation across the population of social media users, not to identify expert emoji users. Including participants with varying levels of familiarity with emoji, from heavy daily use to rare use, enriches the sample and increases the validity of the findings.

Demographically, the final sample of 70 respondents includes 35 men (50%), 33 women (47.1%), and 2 non-binary individuals (2.9%). Education is skewed towards higher levels: 52.9% hold postgraduate qualifications, 38.6% hold undergraduate qualifications, 5.7% hold secondary school qualifications, and 2.9% hold doctoral qualifications. Linguistically, the sample is highly diverse: primary languages reported include Telugu (n=26), English (n=17), Hindi (n=8), Tamil (n=3), Odia (n=2), Marathi (n=2) and others. This linguistic diversity reflects the study's South Asian recruitment context. It provides a multilingual dimension that is underrepresented in most existing emoji research, which is heavily concentrated in English-speaking Western populations (Barinaga-López, et al., 2022).

3.6.2. Data Sources

The study draws on only one primary data source: the online questionnaire described in Section 3.4. All quantitative and qualitative data presented in the analysis chapter were collected through this instrument. No secondary data sources, archival corpora, or external datasets were included in the analysis. This single-source design was chosen to ensure consistency of measurement across all variables and to maintain the study's focus on self-reported interpretation and use rather than on naturally occurring emoji production, which would require a fundamentally different data-collection approach.

The questionnaire generated data of three types. The first type is ordinal Likert-scale data from the nine perception statements in Block 4 and five usage frequency items in Block 3. The second type is categorical data from the demographic questions in Block 1, the contextualised interpretation tasks in Block 3, and the yes/no questions in Block 5. These data were exported directly from Google Forms as an Excel file and loaded into a pandas DataFrame[7] for analysis. The third type of data is the unstructured qualitative data from the open-ended questions in Blocks 3 and 5, which asked respondents to explain their interpretive choices in their own words. These responses were exported as text and analysed using a manual coding process described in Section 3.6.4.

Google Forms was selected as the data-collection medium for several reasons. It is freely accessible and requires no special software installation on the respondent's device, reducing participation barriers. It renders emoji as expected by the researcher in the survey interface, ensuring respondents see the emoji as intended rather than as character codes for replacement glyphs. It supports conditional routing, which allows the questionnaire to present follow-up questions only to respondents who indicated familiarity with a given emoji. It produces a clean

[7] A pandas DataFrame is a two-dimensional, size-mutable data structure with labeled axes (rows and columns) designed for efficient data manipulation and analysis in Python

structured data export in Excel format that is compatible with the pandas analytical pipeline. It also stores responses in a secure cloud environment that is accessible to authorised account holders, satisfying the anonymity requirements described in Section 3.5.

The data source is entirely self-reported. Respondents described their emoji use and interpretation, rather than being observed using them in naturalistic communication environments. This design choice prioritises breadth (the ability to collect systematic data from 70 participants across four cohorts on multiple variables) over the ecological validity that would be achieved through analysis of naturally occurring WhatsApp or Instagram message corpora. The trade-offs of the self-report methodology are discussed again in Section 3.6.5. The contextualised interpretation tasks in Block 3, which present respondents with specific emoji-in-sentence stimuli and ask for interpretive judgements, partially mitigate this limitation by capturing online interpretive processes rather than retrospective self-assessment.

3.6.3. Data Collection Procedure

Data collection lasted approximately three weeks. The questionnaire link was first shared via personal social media accounts, accompanied by a brief description of the study. The snowball element of this distribution was important for reaching the 35-45 and 45+ cohorts, who are less likely to participate without the researcher's personal network. In the second week, the questionnaire was also shared on social media platforms, which generated additional responses primarily from the 18-25 cohort. In the third week, targeted outreach was conducted through professional contacts to increase representation of older cohorts, and the questionnaire was shared directly via messaging forums with personal messages for those groups.

Respondents completed the questionnaire independently, and it was administered online, at a time and place of their choosing. No researcher was present during completion, and no time limit was imposed. The median completion time, estimated from the Google Forms response timestamps, was approximately twelve to fifteen minutes. This duration is appropriate for the instrument's length and complexity: long enough to elicit thoughtful responses to the open-ended questions but short enough to avoid respondent fatigue, which can compromise data quality in longer surveys.

Respondents were not compensated for their participation. Participation was voluntary and driven by social connection to the researcher or members of the researcher's distribution network. This non-compensated design is well-suited for a dissertation study of this scale and avoids the demand characteristics and response biases that can accompany incentivised participation. It is acknowledged that voluntary, unpaid participation tends to select participants who are motivated by social connection to the researcher or by genuine interest in the topic, potentially introducing self-selection bias. This is discussed further in Section 3.6.5.

3.6.4. Data Preparation and Cleaning

Raw data was exported from Google Forms as an Excel file containing seventy rows (one per respondent) and forty-four columns (one per survey question). The dataset was loaded into a Python pandas environment for inspection and cleaning. The first analytical step was a structural inspection of the dataset: examination of column names, data types, unique value distributions, and missing data patterns. This inspection did reveal some unused columns, but no empty rows were identified, and all 70 respondents completed the questionnaire in full.

Demographic variables were extracted and verified. The age column contained four unique values corresponding to the four survey brackets, with no missing values. Gender contained three categories (male, female, non-binary), and was also complete. Education was categorised into four groups with no missing values. The age groups were confirmed to map cleanly to the study's four analytical cohorts without requiring recording, since the survey brackets aligned with the theoretical distinctions described in Section 3.3.

Emoji usage frequency columns were identified by their question text and verified against the expected Likert scale of 1-5. Value distributions were inspected for each of the five-target emoji to confirm the responses fell within the valid range and that the mapping of numeric values to frequency labels (1=Never, 2=Rarely, 3=Sometimes, 4=Often, 5=Very Often) was internally consistent. No out-of-range values were identified. Cross-tabulations were computed using pd.crosstab() for each emoji against the age group variable, producing raw count tables that were subsequently converted to row-normalised percentage tables. Row normalisation was chosen over column normalisation because it enables a fair comparison of each cohort's distribution across frequency categories, independent of each cohort's absolute size.

Qualitative open-ended responses were subjected to a multi-step manual coding process. All responses were first read in full to identify the range of semantic themes present. This initial reading produced a preliminary list of recurring categories that was refined into a formal coding frame. The coding frame for the contextualised interpretation tasks distinguished between ironic/sarcastic readings, literal/sincere readings, ambivalent or unclear readings, and absent or non-engaged responses ("don't know" or blank). The coding frame for the open-ended "what do you mean when you send this emoji" questions distinguished between humour/laughter, sarcasm/irony, literal emotion (sadness, danger), aesthetic or decorative use, and context-dependent or ambivalent use. Each response was assigned to the most appropriate category, and a second reading of the full coded dataset was conducted to verify consistency. Where the assignment was ambiguous, a conservative coding was applied (typically ambiguous) to avoid overcounting ironic or literal readings.

Derived metrics were computed from the cleaned and coded dataset. Active usage was defined as the percentage of each cohort reporting Often (4) or Very Often (5) on the usage frequency scale. Passive usage was defined as the percentage reporting Never (1) or Rarely (2). The sarcasm classification variable (high sarcasm 4-5; low sarcasm 1-2) was computed from the Likert scale sarcasm item. Decline rates across cohorts were computed as the absolute percentage-point difference in active usage between the 18-25 and 25-35 cohorts, expressed both as an absolute difference and as a relative percentage change. Validation checks were performed to ensure that row sums in all percentage tables equalled 100%, that no cells contained negative values, and that extreme percentages (e.g., 93.8% of 35–45 reporting never using the skull emoji) were consistent with the raw count data rather than being artefacts of computation.

3.6.5. Limitations of the Data Collection Process

The most important limitation is the sample size. A total of n=70 respondents, distributed across four age cohorts with cell sizes ranging from 9 to 28, limits the statistical power that is available for between-cohort comparisons. Small cells, particularly the 45+ cohort (n=9), make percentages within that group difficult to interpret, since each response accounts for approximately 11% of the group total. Cross-tabulation and descriptive statistics at this scale can identify strong patterns with confidence but cannot reliably detect smaller or more nuanced effects. The study, therefore, presents its findings as indicative of real structural differences rather than as statistically definitive.

A second limitation is the sampling method. Convenience sampling through personal networks introduces systematic biases: the sample is skewed towards higher education (91.5% of respondents at the undergraduate level or above); younger age (40% from the 18-25 age cohort); and digital literacy. The characteristics limit the generalizability of findings. In particular, the higher-education skew may lead the study to underestimate the difficulties faced by less digitally literate populations.

The third is the cross-sectional design's inability to disentangle age effects from cohort effects. The study observes differences between 18-25 and 45+ users at a single point in time. Still, it cannot determine whether those differences reflect generational socialisation (cohort effect: users interpret differently because they encountered emoji at different historical moments) or current life stage (age effect: users interpret differently because of their current life stage and social context). Cohort and age are necessarily confounded in any cross-sectional design. A longitudinal study tracking the same individuals over time, or a natural experiment exploiting a historical discontinuity in emoji adoption, would be required to separate these effects. This limitation is shared by virtually all empirical studies of generational differences in digital communication.

Finally, the last limitation concerns the validity of respondents' self-reports. The questionnaire asks respondents to report their emoji behaviour and interpretation preferences, but it does not observe those behaviours in naturalistic communication. Self-reported behaviour may diverge from actual behaviour in several systematic ways. For example, respondents may report more ironic or sophisticated emoji use than they actually deploy (social desirability bias), may not be consciously aware of their own interpretive defaults (metacognitive limitations), and may respond differently in the questionnaire context than they would in real conversational contexts (demand characteristics). The contextualised interpretation tasks in Block 3 partially mitigate this concern by presenting concrete stimuli and asking for specific judgements rather than general self-assessments. Still, they remain a laboratory-style measure rather than a naturalistic one.

Chapter 4
Data Analysis and Discussion

4.1. Overview of Collected Data

The data for this study were collected via an online questionnaire. This analysis examines seventy respondents across four age cohorts (18-25, 25-35, 35-45, 45+) to identify patterns in the semantics, pragmatics, and interpretation of emoji. The respondents answered forty-four questions, ranging from emoji interpretation tasks to scalar items and other open-ended prompts, designed to elicit understandings of emoji in both broad and specific senses. The data reveals significant differences in emoji meaning-making by age, with younger users adopting a more ironic/contextual framework while older users maintain literal interpretations.

Sample Composition (n=70)

Table 1.1: Demographic description of respondents

Dimension	Category	N	%
Age Group	18-25	28	40.0%
	25-35	17	24.3%
	35-45	16	22.9%
	45+	9	12.9%
Gender	Male	35	50.0%
	Female	33	47.1%
	Non-binary	2	2.9%
Education	Postgraduate	37	52.9%
	Undergraduate	27	38.6%
	Secondary (10-12)	4	5.7%
	PhD/Graduate	2	2.9%

The study is leaning toward the younger demographic, with respondents aged 18-35 accounting for 64.3%. The sample also leans towards higher education, with 91.5% of respondents holding undergraduate qualifications or higher. The gender balance is reasonable, with only a slight male majority. The sample is highly linguistically diverse, with primary languages being Telugu (n=26), English (n=17), Hindi (n=8), Tamil (n=3), Odia (n=2), Marathi (n=2), and others. Despite several regional languages being spoken as primary languages, English is the primary language of social media use across the sample, supporting the decision to administer the questionnaire in English.

This sample reflects a highly educated, digitally literate population. It may not fully represent the less educated or less digitally engaged populations.

4.2. Frequency and Distribution of Emoji Usage

Table 1.2: Overall Emoji Usage Frequency

Frequency Level	Count	Percentage
Very Often	14	20%
Often	32	45.7%
Sometimes	13	18.6%
Rarely	10	14.3%
Never	1	1.4%

A clear majority shows that 65.7% of respondents use emoji "often" or "very often", indicating that emoji are integral to digital communication, according to the respondents.

Table 1.3: Distribution of Emoji Usage Frequency by Age Group

Age Group	Never	Rarely	Sometimes	Often	Very Often
18-25	1 (3.6%)	0 (0.0%)	5 (17.9%)	15 (53.6%)	7 (25.0%)
25-35	0 (0.0%)	4 (23.5%)	4 (23.5%)	3 (17.6%)	6 (35.3%)
35-45	0 (0.0%)	3 (18.8%)	1 (6.2%)	11 (68.8%)	1 (6.2%)
45+	0 (0.0%)	3 (33.3%)	3 (33.3%)	3 (33.3%)	0 (0.0%)

Table 1.3 shows that active emoji usage is highest among the 18-25 cohort, where 22 out of 28 respondents (78.6%) report using emoji Often or Very Often. The 35-45 cohort also reports high active use, with 12 out of 16 respondents (75.0%) selecting Often or Very OFten. The 25-35 age group shows a more moderate pattern, with 52.9% reporting active use, while the 45+ cohort reports the lowest active usage rate at 33.3%. These results suggest that emoji use is not limited to the youngest respondents, although the youngest cohort remains the most consistently active group. At the same time, the comparatively low active-usage rate among the 45+ cohort indicates that age continues to shape the frequency of emoji use in everyday digital communication.

This pattern shows that emoji use is well established across the segment of the population that engages with online media. It also makes an important distinction for the present study: age alone is not a strong predictor of emoji use frequency. Even older demographic groups show substantial engagement. Differences in interpretation, rather than in raw usage, are where the generational divergences will appear.

Table 1.4: Social Media Platform Preferences (Primary)

Age Group	Platform 1 (%)	Platform 2 (%)
18-25	Instagram (71%)	WhatsApp (18%)
25-35	WhatsApp(53%)	Instagram (41%)
35-45	WhatsApp (44%)	Instagram (38%)
45+	WhatsApp (78%)	Instagram (22%)

A clear generational shift in platform preference is observed: younger users prefer Instagram, while older users prefer WhatsApp.

Table 1.5: Daily Social Media Usage Hours by Age

Age Group	<1 Hour	1-2 Hours	3-4 Hours	5+ Hours
18-25	2 (7.1%)	10 (35.7%)	12 (42.9%)	4 (14.3%)
25-35	4 (23.5%)	6 (35.3%)	5 (29.4%)	2 (11.8%)
35-45	2 (12.5%)	6 (37.5%)	7 (43.8%)	1 (6.2%)
45+	2 (22.2%)	6 (66.7%)	1 (11.1%)	0 (0.0%)

Daily social media usage varies across age cohorts. The 18-25 cohort reports the highest proportion of heavy social media use, with 16 out of 28 respondents (57.2%) using social media for three or more hours per day. The 35-45 cohort also shows relatively high engagement, with 8 out of 16 respondents (50.0%) reporting three or more hours per day. The 25-35 cohort is more moderate, with 7 out of 17 respondents (41.2%) in the three-or-more-hours category. In contrast, the 45+ cohort is concentrated primarily in the 1-2 hour category, with 6 out of 9 respondents (66.7%) selecting this option and no respondents reporting 5+ hours of use. This suggests that younger respondents are more likely to report heavier daily social media engagement, although moderate-to-high usage is also present among the 35-45 cohort.

4.3. Semantic Variations in Emoji Interpretation

4.3.1. The Skull Emoji 💀

Table 1.6: Usage Frequency of 💀 by Age

Age Group	Never	Rarely	Sometimes	Often	Very Often
18-25	21.4%	10.7%	10.7%	39.3%	17.9%
25-35	58.8%	17.6%	5.9%	5.9%	11.8%

35-45	93.8%	-	6.2%	-	-
45+	77.8%	-	22.2%	-	-

The pattern is as expected: younger users (18-25) use the skull emoji frequently, with active usage of 57.2% (Often + Very Often), while users aged 35 and over use it rarely or not at all (active usage of 0% in the 35-45 cohort and 0% in the 45+ cohort). This is the most age-differentiated emoji in the study.

Contextual Interpretation: "That presentation was amazing 💀"

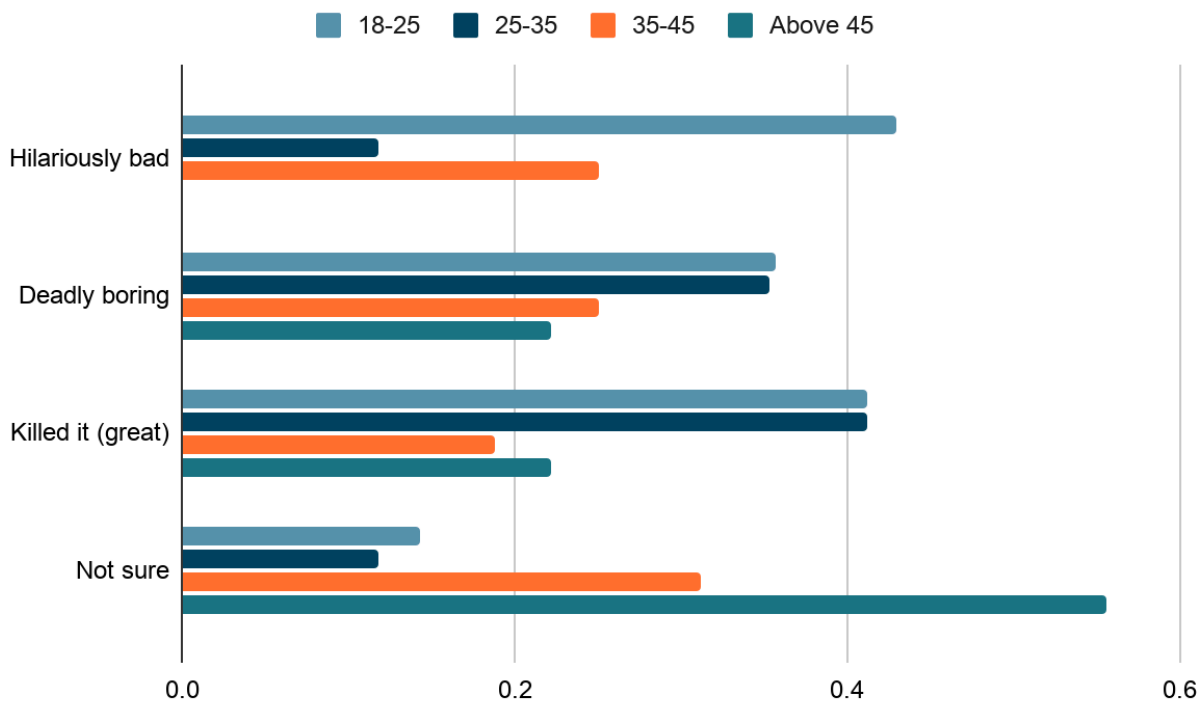

Figure 1.1: Contextual interpretation of 💀 among all age cohorts

The 💀 emoji exhibits high semantic variation across age groups, with younger users, as expected, more likely to interpret it as humour.

The 18-25 group shows a majority understanding that the 💀 emoji conveys humour, with 45% agreeing. From this data, however, we can see that this age group does use the emoji to mean something close to death, as two options with death-related wording show a high usage percentage. A small minority of the group claims to be unsure about the emoji's use in this context.

The 25-35 age group is in a transitional phase, where there is almost equal use of the emoji across all senses marked in the questionnaire. Surprisingly, the percentage of respondents who claim "not sure" is lower than that of the 18-25 age group.

We can only note the change in the next two age groups (35-45 and 45+). Here, there is a marked difference in how respondents in this age group understand the 💀 emoji, with more respondents marking "unsure." This number goes up with each age group.

4.3.2. 😭(Loudly Crying Face Emoji)

Table 1.7: Usage Frequency of 😭 by Age

Age Group	Never	Rarely	Sometimes	Often	Very Often
18-25	3.6%	-	14.3%	25.0%	57.1%
25-35	11.8%	17.6%	35.3%	17.6%	17.6%
35-45	12.5%	25.0%	43.8%	18.8%	-
45+	-	33.3%	55.6%	11.1%	-

The 😭 emoji shows a far more balanced age distribution in terms of usage compared to the 💀 emoji. Younger users use it far more frequently (57% "Very Often"), and older users still maintain regular usage (44% "Sometimes").

Contextual Interpretation: "Stop 😭"

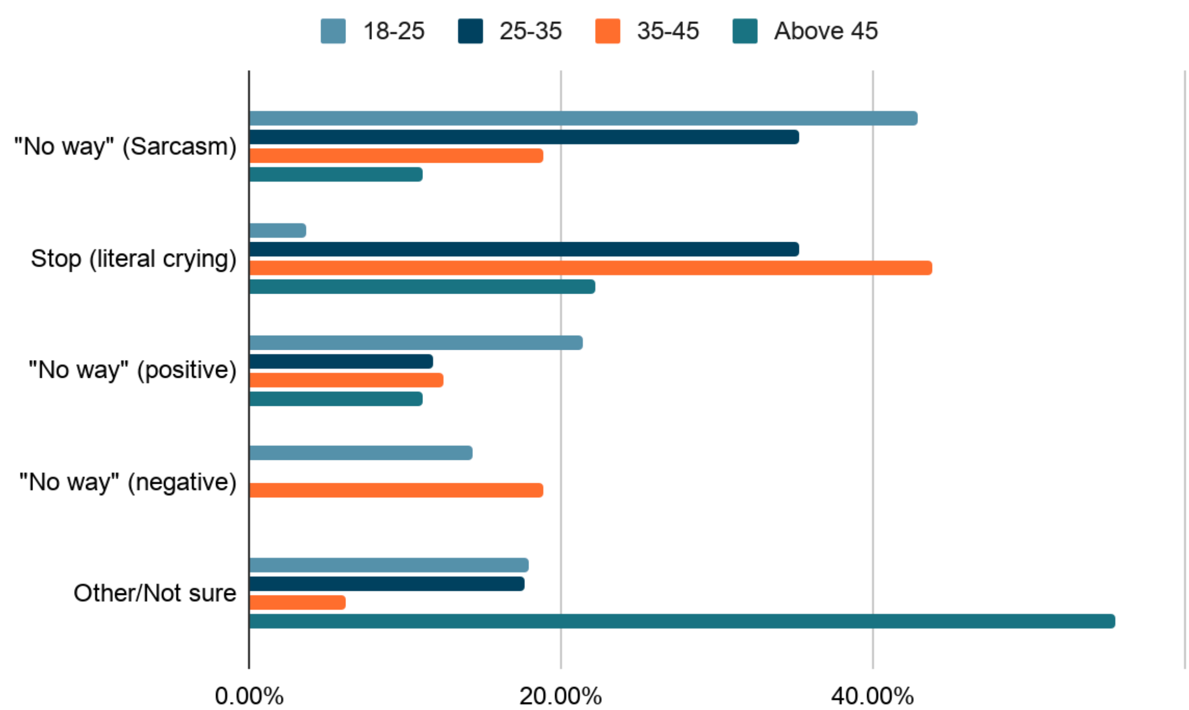

Figure 1.2: Contextual interpretation of 😭 across all age cohorts

Interpretation diverges sharply by age. In the 18–25 cohort, 42.9% read the construction as sarcastic denial — the "no way!" reading. The 25–35 cohort splits between the sarcastic denial and literal crying readings, both at about 35.3%. In the 35–45 cohort, 43.8% read it as literal sadness or distress, with sarcastic denial at only 18.8%. In the 45+ cohort, only 11.1% recognise sarcasm.

This pattern has direct implications for cross-generational communication. A young user sending "stop 😭" to an older relative as a sarcastic protest may be read as genuinely upset, producing emotional concern that the sender did not intend. The same construction can therefore generate confusion in either direction, depending on which cohort is sending and which is receiving.

4.3.3. ✨(Sparkles Emoji)

Table 1.8: Usage Frequency of ✨ by Age

Age Group	Never	Rarely	Sometimes	Often	Very Often
18-25	10.7%	17.9%	32.1%	14.3%	25.0%
25-35	29.4%	23.5%	35.3%	5.9%	5.9%
35-45	43.8%	12.5%	31.2%	-	12.5%
45+	55.6%	11.1%	33.3%	-	-

The ✨ emoji is already rarely used in this sample as a whole, and its usage declines with age. Active usage drops from 39.3% in the 18-25 cohort to 11.8% in the 25-35 cohort, 12.5% in the 35-45 cohort, and 0% in the 45+ cohort.

Contextual Interpretation: "Just finished my taxes ✨"

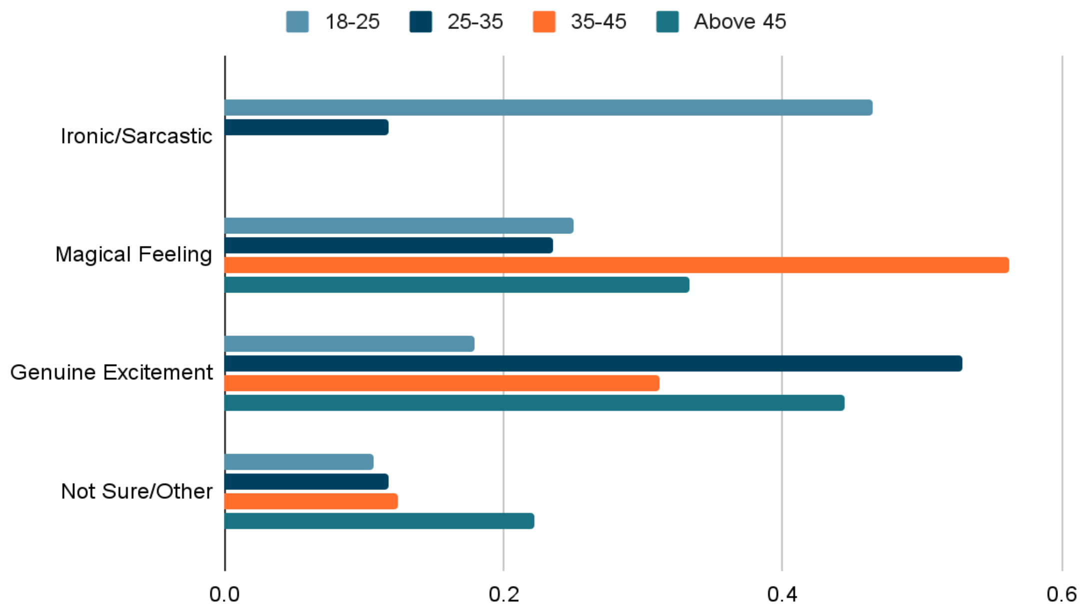

Figure 1.3: Contextual interpretation of ✨ among all age cohorts

There is a shift across age groups from sarcasm to sincerity. In the 18-25 cohort, 46.6% interpret the ✨ emoji as ironic– taxes, after all, are not magical– while smaller portions read it as genuine excitement (17.9%) or as a "magical feeling" marker (around 25%). The 25-35 cohort shows an upward trend toward sincerity, with 52.9% reading the construction as genuine excitement. The 35–45 cohort interprets the sparkles primarily as a marker of a magical or aesthetic feeling (56.2%). The 45+ cohort largely reads it as genuine excitement (44.4%) or a magical feeling (33.3%).

The ✨ emoji is more forgiving than the skull or crying emoji, in the sense that even younger users accept the sincere reading as valid alongside the ironic one. However, only respondents under 35 select the ironic readings; respondents over 35 do not assign sarcasm to the ✨ in this construction.

4.3.4. 🙂(Slightly Smiling Face Emoji)

Table 1.9: Usage Frequency of 🙂by Age:

Age Group	Never	Rarely	Sometimes	Often	Very Often
18-25	3 (10.7%)	8 (28.6%)	5 (17.9%)	8 (28.6%)	4 (14.3%)
25-35	4 (23.5%)	2 (11.8%)	4 (23.5%)	4 (23.5%)	3 (17.6%)
35-45	1 (6.2%)	1 (6.2%)	4 (25.0%)	3 (18.8%)	7 (43.8%)
45+	0 (0.0%)	3 (33.3%)	1 (11.1%)	1 (11.1%)	4 (44.4)

The slightly smiling face emoji is used across all age cohorts, but its frequency is not uniform. Active use is highest among the 35-45 cohort, where 10 out of 16 respondents (62.5%) report using it Often or Very Often, followed by the 45+ cohort at 55.6%. The 18-25 and 25-35 cohorts show lower active-use rates, at 42.9% and 41.2% respectively. This pattern is particularly important because the slightly smiling face emoji is often pragmatically reinterpreted by younger users as passive-aggressive, awkward, or ironic. The frequency data therefore suggest that shared use does not necessarily imply shared meaning: older cohorts may use the emoji more literally or positively, while younger cohorts may attach more indirect or context-dependent meanings to it.

Contextual Interpretation: "Thanks for sending that report 🙂"

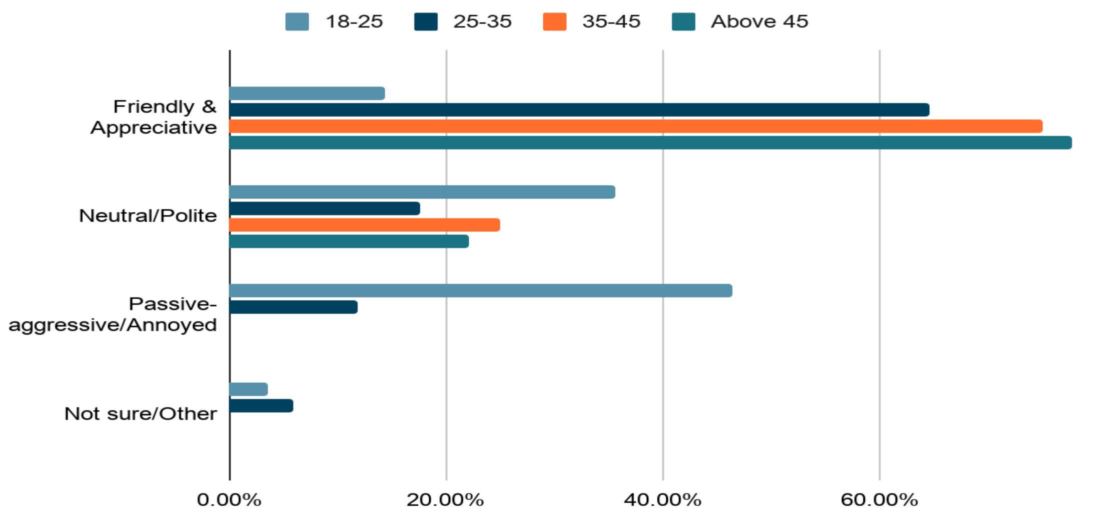

Figure 1.4: Contextual interpretation of 🙂 among all age cohorts

This emoji yields the most striking interpretive divergence in the entire dataset. In the 18-25 cohort, 46.4% of the respondents read the construction as passive-aggressive or annoyed. In the 25-35 cohort, the interpretation shifts dramatically: 64.7% read it as friendly and appreciative. The shift continues in the 35-45 cohort, where 75.0% read it as friendly and appreciative, and culminates in the 45+ cohort, where 77.8% interpret it as friendly and appreciative.

This reversal, where the youngest cohort defaults to a passive-aggressive reading and the oldest cohort defaults to a friendly reading of the same emoji in the same sentence, is one of the clearest illustrations in the dataset of how the same symbol can simultaneously perform different communicative acts depending on the interpretive frame the receiver brings to it. The implications for workplace and family communication are direct.

4.3.5.🥀(Wilted Flower Emoji)

Table 1.10: Usage Frequency of 🥀 by Age:

Age Group	Never	Rarely	Sometimes	Often	Very Often
18-25	14 (50.0%)	6 (21.4%)	5 (17.9%)	0 (0.0%)	3 (10.7%)

25-35	15 (88.2%)	1 (5.9%)	1 (5.9%)	0 (0.0%)	0 (0.0%)
35-45	13 (81.2%)	2 (12.5%)	1 (6.2%)	0 (0.0%)	0 (0.0%)
45+	9 (100.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

The wilted flower emoji is a low-frequency emoji across the sample. Even in the 18-25 cohort, only 3 out of 28 respondents (10.7%) report active use, while half of the cohort reports never using it. The pattern is even stronger in the older cohorts: 88.2% of 25-35 respondents, 81.2% of 35-45 respondents, and all 45+ respondents report never using the emoji. This suggests that the wilted flower functions less as a widely used everyday emoji and more as a niche or subculturally marked sign. Its relevance to the study therefore lies in how respondents interpret a less frequent but potentially generation-linked pragmatic marker.

Contextual Interpretation: "He thinks he's cool like that 🥀"

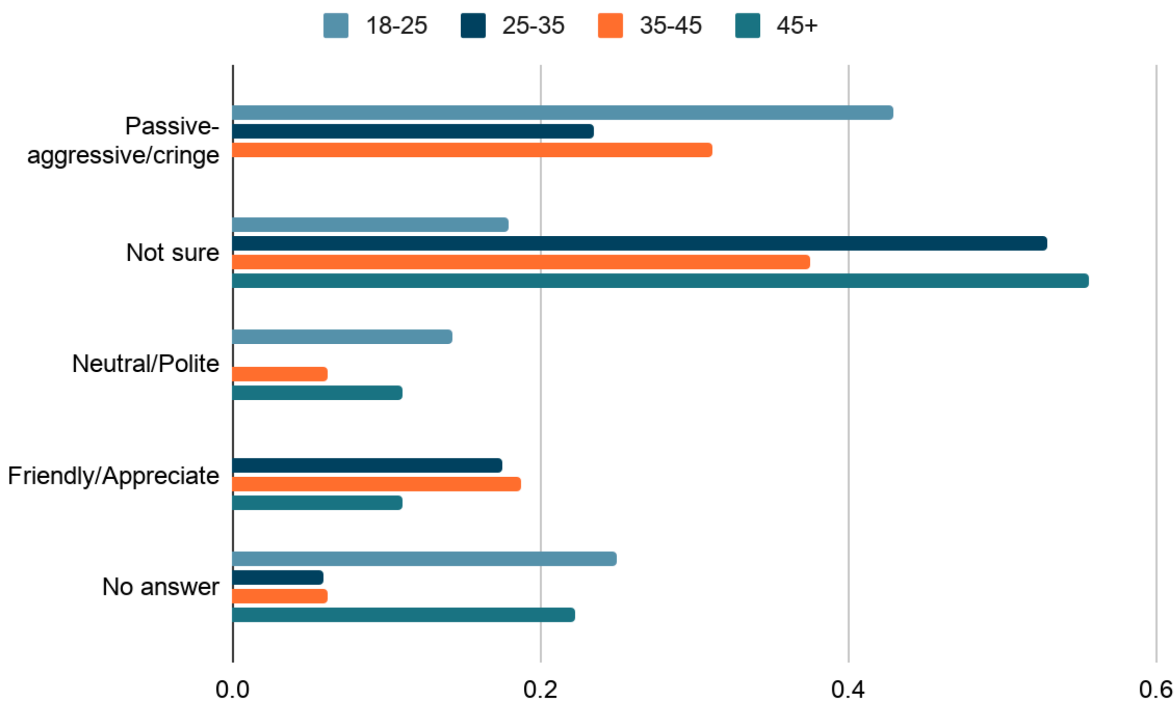

Figure 1.5: Contextual interpretation of 🥀 among all age cohorts

This emoji reveals a generational knowledge disparity. The 18-25 cohort is most familiar with it: 42.9% interpret it as marking sarcasm or "cringe". The 25-35 cohort is largely unsure, with 52.9% selecting "don't know". The 35-45 cohort splits between confusion (37.5% "don't know") and recognition of the sarcasm reading (around 31%). The 45+ cohort is overwhelmingly unsure (55.6% "Don't Know").

The 🥀 emoji functions in this dataset as a marker of bottom-up generational innovation. Young users have developed a new sarcasm code (wilted = dying = cringe = pathetic) that has not spread to older demographics. Open-ended responses to these questions reflect this. Several respondents from older cohorts wrote variants of “the wilted rose is a younger people thing”, and some reported that older relatives had asked for explanations of the emoji’s meaning.

The wilted flower contextual item received 10 missing responses across the sample. Therefore, interpretation of this item should be treated cautiously. In reporting the results, passive-aggressive/annoyed selections and the open response "cringe" were treated as sarcasm-like or negative pragmatic readings because both move away from the literal botanical meaning of the emoji and toward an evaluative stance toward the speaker or message.

For the wilted flower contextual item, the 18-25 cohort showed the strongest tendency toward a negative or sarcasm-like interpretation. Twelve respondents in this cohort selected the passive-aggressive/annoyed reading, and one additional respondent supplied "cringe" as an open response. If these are combined as sarcasm/cringe readings, the total becomes 13 out of 28 respondents (46.4%). The corresponding combined rates were 23.5% for the 25-35 cohort and 31.2% for the 35-45 cohort, while no 45+ respondents selected or supplied this interpretation. However, because the item had 10 missing responses overall, these findings should be interpreted as indicative rather than conclusive.

Across the five emoji, the 18-25 and 25-35 cohorts strongly resist fixed meanings in favour of contextual interpretations. This represents a paradigm shift in how emoji are semantically processed. For younger users, emoji function as polysemous signs whose denotation is subordinate to pragmatic context. For older users, emoji maintain a closer indexical or iconic correspondence to the emotion or the object depicted, producing more literal readings.

4.4. User Perception Patterns

This section will focus on respondents’ perceptions of emoji and the patterns in those perceptions. The perceptions examined include: tone clarification, use of words instead, changes in meaning across platforms, stability of meaning, the possibility and experience of misunderstandings, sarcastic usage, and context-dependency.
The information has been sorted by the number of people in each age group who agree with the questionnaire statements, and responses are measured on a Likert scale.

Tone Clarification
I feel that emoji help clarify the tone in text-based communication.

Table 1.11: Tone clarification agreement across age cohorts (1)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	1	0	3	13	11
25-35	0	2	3	2	10
35-45	2	0	3	3	8

45+	0	0	5	2	2

A clear majority strongly believes that emoji clarify tone, indicating consensus on their utility as a communicative resource.

Intensify Emotion
Statement: I use emoji to intensify emotion

Table 1.12: Tone clarification agreement across age cohorts (2)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	4	2	2	9	11
25-35	1	2	4	4	6
35-45	3	2	6	3	2
45+	2	1	3	1	2

Across the sample, 54.3% of respondents use emoji to amplify emotional expression, a pattern that is broadly consistent across ages, although the youngest cohort is most enthusiastic.

Soften Criticism
Statement: I use emoji to soften criticism

Table 1.13: Tone clarification agreement across age cohorts (3)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	3	3	5	9	8
25-35	3	2	5	2	5
35-45	2	8	5	0	1
45+	4	2	3	0	0

Around 35.7% of respondents regularly use emoji to soften criticism. Age differences are present but less pronounced than for sarcasm, with younger cohorts more likely to use emoji for face-management functions.

Use Sarcastically

Statement: I use emoji sarcastically

Table 1.14: Tone clarification agreement across age cohorts (4)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	3	1	2	12	10
25-35	2	1	5	1	8
35-45	4	6	3	0	3
45+	4	2	1	0	2

This statement reveals a substantial generational gap. Active sarcasm use ("Agree" + "Strongly Agree") is reported by 78.6% of the 18-25 cohort. This difference largely explains the divergent interpretation patterns documented in Section 4.3: cohorts that habitually deploy sarcasm through emoji are also the cohorts most likely to recognise others' sarcastic intent in emoji.

Use Instead of Words
Question: I sometimes use emoji instead of words

Table 1.15: Tone clarification agreement across age cohorts (5)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	1	3	2	11	11
25-35	1	1	4	4	7
35-45	4	1	2	3	6
45+	1	1	2	3	2

Across the sample, 67.1% of respondents use emoji as substitutes for words. This indicates that emoji are functioning as a substitute lexical item rather than as supplementary decoration in a substantial portion of digital communication.

Meaning Change Over Time
Statement: Emoji meanings have changed over time since I started using them.

Table 1.16: Tone clarification agreement across age cohorts (6)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	2 (7.1%)	0 (0.0%)	3 (10.7%)	11 (39.3%)	12 (42.9%)
25-35	1 (5.9%)	3 (17.6%)	7 (41.2%)	1 (5.9%)	5 (29.4%)

35-45	2 (12.5%)	4 (25.0%)	4 (25.0%)	1 (6.2%)	5 (31.2%)
45+	1 (11.1%)	1 (11.1%)	1 (11.1%)	3 (33.3%)	3 (33.3%)

The 18-25 cohort has the strongest agreement that emoji meanings have changed over time, with 23 out of 28 respondents (82.1%) selecting Agree or Strongly Agree. The 25-35 and 35-45 cohorts show lower agreement, at 35.3% and 37.5% respectively, with comparatively higher neutrality or disagreement. The 45+ cohort also shows relatively high agreement at 66.7%, although this should be interpreted cautiously because the cohort contains only nine respondents. Overall, the table suggests that awareness of semantic change is strongest among younger respondents, but it is not absent among older users

Context-Dependent
Statement: Emoji meaning depends heavily on context

Table 1.17: Tone clarification agreement across age cohorts (7)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	1	0	5	14	8
25-35	0	1	2	7	7
35-45	3	1	5	5	2
45+	0	1	2	4	2

Around 70% of all respondents either strongly agree or agree that context is crucial to emoji meaning. This is the highest consensus belief across all the perception statements.

Stable/Fixed Meaning
Statement: Emoji have stable, fixed meanings

Table 1.18: Tone clarification agreement across age cohorts (8)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	6	8	11	2	1
25-35	0	3	8	4	2
35-45	1	0	3	6	6
45+	0	3	2	2	2

Approximately 30% of respondents agree that emoji have fixed meanings, and another 30% disagree. This is a polarised distribution, suggesting different underlying philosophies of emoji semantics: the youngest cohort shows the highest disagreement, and the 35-45 cohort shows the highest agreement.

Meaning Changes by Platform
Statement: Emoji meaning changes depending on platform

Table 1.19: Tone clarification agreement across age cohorts (9)

Age Group	Strongly Disagree (1)	Disagree (2)	Neutral (3)	Agree (4)	Strongly Agree (5)
18-25	6	4	7	5	6
25-35	5	1	7	2	2
35-45	8	1	3	2	2
45+	2	0	3	3	1

Here, an unexpected pattern emerges: the youngest and oldest cohorts tend to agree that the platform affects meaning, while the middle cohorts are more sceptical, with the highest rates of neutrality.

Experienced Misunderstanding
Statement: I have misunderstood an emoji before

Table 1.20: Misunderstanding of emoji across age cohorts

Age Group	Yes	No	Maybe
18-25	16	4	8
25-35	6	6	5
35-45	7	9	0
45+	7	2	0

Across the full sample, 51.4% of respondents (36 of 70) report having misunderstood an emoji at some point. The finding validates the concerns about semantic instability and ambiguity raised throughout the analysis. Given that 65.7% of the sample reports using emoji often or very often, the rate of misunderstanding is striking: roughly half of regular emoji users have experienced communicative breakdown around an emoji.

4.5 Key Observations Related to Research Questions

4.5.1. The Great 💀 Divide: Sarcasm as a Generational Boundary

The 💀(skull) emoji emerged as the most age-differentiated emoji in the study. The 18-25 cohort shows 57.2% active usage with the dominant interpretation that the 💀 marks humour, a clearly pragmatically reanalysed reading that is consistent with the "laugh till I die" usage documented elsewhere in the literature. By contrast, 93.8% of the 35-45 cohort and 77.8% of the 45+ cohort never use it; those who do interpret it more literally, with the "killed it" reading retaining its straightforward death-as-success connotation. The skull, in this dataset, encodes a generational code of humour. What 18-25-year-olds deploy to mark amusement is interpreted literally (or not at all) by users over 35.

4.5.2. Instagram Vs. WhatsApp: The META Battle

There is a clear preference for which platform each cohort gravitates toward. The 18-25 cohort skews heavily toward Instagram (71%), a more visual, ephemeral, peer-based environment. The 45+ cohort skews equally toward WhatsApp (78%), a more text-based, persistent, family-focused environment. This platform difference may itself shape the divergent emoji semantics observed in the study, as the conventions that govern emoji use on Instagram differ from those on WhatsApp.

4.5.3. Context Trumps All

The single highest-agreement statement in the entire questionnaire was "Emoji meaning depends heavily on context," with 70% of respondents agreeing or strongly agreeing. Yet only about 40% report that, in their experience, emoji meanings shift in the contextual examples provided. This gap suggests that respondents understand context-dependence intellectually but may not apply it consistently in practice. The implication is that interpretive defaults (likely cohort-specific) override the abstract recognition that context matters.

4.5.4. Emoji Chaos: A Universal Struggle

The 51.4% misunderstanding rate observed in the sample suggests that approximately one in two regular emoji users has experienced a communicative breakdown around an emoji. Given the rapidity of semantic change documented in the literature (Robertson, et al., 2021; Schneebeli, 2025), this rate is unsurprising: emoji meanings are evolving at a pace that outstrips the ability of cross-generational communicators to keep up. The result is a communicative system that is highly expressive within cohorts but unstable across them.

4.5.5. Sarcasm as the Driver of Generational Divergence

The most striking generational asymmetry in the dataset is the use of sarcasm; around 79% of users in the 18-25 cohort report using emoji for sarcasm, compared to about 22% in the 45+ cohort. This is the structural feature that explains the divergent interpretation patterns observed for the skull, the slightly smiling face, and the wilted flower. Users who use emoji sarcastically expect to receive sarcasm; users who do not do not. The cohort that adopted emoji during the 2012-2016 irony-adoption window has internalised sarcasm as a default register, while older cohorts retain more transparent, iconic interpretations. This is the generational fault line on which the rest of the divergences rest.

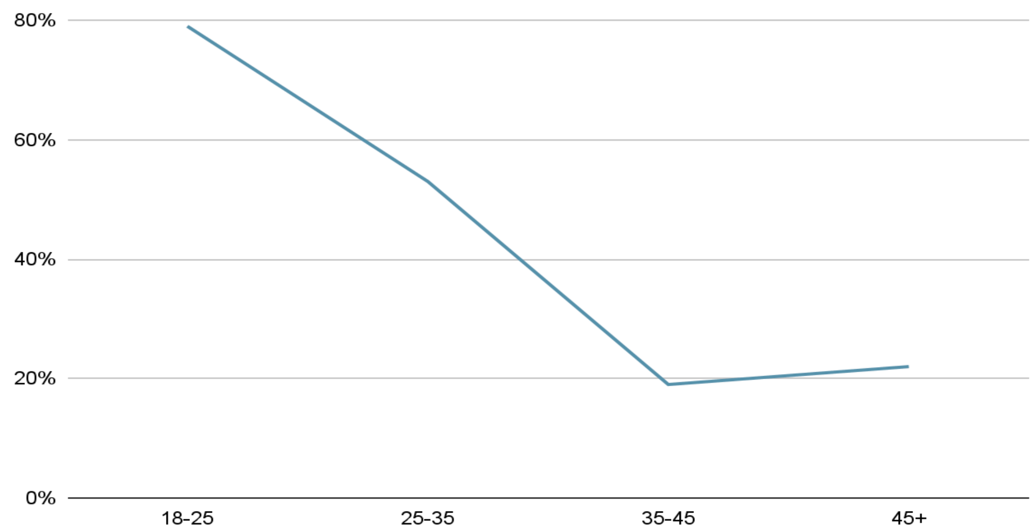

Figure 1.6: Usage of sarcasm through emoji across all age cohorts

Chapter 5
Conclusion

5.1. Summary of Key Findings

This study examined age-based differences in the interpretation of emoji across four age cohorts (18–25, 25–35, 35–45, and 45+) using a survey with 70 respondents. The central findings can be organised around observations made throughout the study that, when taken together, constitute a coherent and theoretically grounded account of the divergence in emoji use.

Finding 1: The 25-35 cohort is a transitional bridge group with a distinct interpretive profile

The single most striking pattern in the data is the emergence of the 25-35 cohort as a demonstrably transitional group. This bridge cohort does not align cleanly with either the youngest or the older cohorts. This cohort shows the highest rates of interpretive uncertainty across multiple target emoji, peaking at 52.9% “don’t know” for the wilted flower (🥀), compared with around 18% in the 18-25 group and similarly low rates in the 35-45 group. It also exhibits the sharpest single-step decline in active emoji usage across most age-differentiated emoji: active skull usage drops by 39.5 percentage points (from 57.2% to 17.7%) between the 18-25 and 25-35 cohorts, a drop larger than any other inter-cohort change. Sarcasm use also declines sharply between these cohorts, from 78.6% to 52.9%, a decline of approximately 26 points that is again larger than any other inter-cohort difference.

The bridge cohort’s pattern of peak ambiguity, sharp decline in usage, and intermediate sarcasm use is consistent with a group undergoing a real-time semantic transition. Members of this cohort have been exposed to the norms of the younger cohort but are increasingly moving away from them. They appear to code-switch between ironic and literal registers, while also exhibiting genuine confusion about which register applies in which context. This finding, more than any other in the study, refines existing accounts of generational emoji divergence by identifying the precise location of the transitional fault line.

Finding 2: Emoji interpretation is systematically stratified by age cohort

The 18-25 cohort interprets the skull (💀), the loudly crying face (😭), and the slightly smiling face (🙂) through a sarcasm-first, pragmatic framework. The skull signals extreme amusement (57.2% active usage; around 43% interpreting “that presentation was amazing 💀” as hilariously bad). The crying emoji signals laughter or sarcasm rather than sadness (42.9% interpreting “stop 😭” as sarcastic denial). The slightly smiling face signals aggression or dismissal (46.4% interpreting “thanks for sending the report 🙂” as passive-aggressive).

The 35-45 and 45+ cohorts reverse these readings across all three emoji. The skull is avoided almost entirely (93.8% of the group never use it) or interpreted literally. The crying emoji is read as genuine sadness (43.8% of the 35-45 group interpret literal crying). The slightly smiling face is read as friendly and appreciative (75.0% of 35–45; 77.8% of 45+). These are not marginal differences; they constitute a near-complete inversion of the semantic frame between the youngest and oldest cohort.

Finding 3: Relationship context outweighs sender age as a predictor of interpretation

Across all four cohorts, between 78% and 88% of respondents report that the sender-receiver relationship affects their emoji interpretation, compared with 67% to 76% who report that sender age affects it. This holds across all cohorts, but the differential weighting of relationship versus age is itself age-stratified, with younger users weighing relationship more heavily. The hierarchical ordering that emerges from the data is as follows: relationship > sender age > platform > message content. This refines the theoretical model by identifying the dominant context variable in emoji interpretation. The practical implication is that an emoji sent by a parent to an adult child, even the same emoji in the same lexical context, will be interpreted differently than one sent by a peer, because the relational framing overrides the ironic coding that would otherwise apply.

Finding 4: Intergenerational miscommunication is widespread and systematically produced

Of the full sample, 51.4% of respondents report having misunderstood an emoji. The qualitative data document specific patterns of cross-generational interpretive failure that are not random but predictable. A parent sends 😭, intending sadness; the adult child perceives humour. An adult child sends 🙂, intending passive-aggressive dismissal; the parent perceives friendly acknowledgement. A young colleague sends 💀 to mark that a meeting was disastrous; the older colleague perceives a literal or alarming connotation. These are not failures of attention or intelligence. They are inevitable communicative outcomes of two parties working with incompatible interpretive systems, as the Level 3 theoretical framework predicts.

5.2. Interpretation of Results in Relation to Research Questions

Research Objective 1: Age-based differences in literal versus pragmatically reanalysed interpretations

The study's findings confirm that the objective was achieved. Different age cohorts diverge in their tendency to assign literal versus pragmatically reanalysed meanings to emoji. This divergence operates at two levels: frequency of use and semantic interpretation.

At the usage level, sarcasm-dependent emoji show sharp declines in active use from the 18-25 cohort to the 25-35 cohort. The skull, crying, and sparkle emoji all decline dramatically, with near-zero active usage in the 35+ cohorts. By contrast, the slightly smiling face shows a flat usage pattern across all groups. This divergence in usage frequency maps directly onto the theoretical distinction between emoji whose usage has remained stable across age cohorts, with no connection to what each cohort uses the same emoji for. The wilted flower emoji, on the other hand, shows very little usage in all the cohorts, and even among those, it is most used by the 18-25 cohort and only shows a sharp decline to near 0 in every cohort after, which can be interpreted as the emoji having just started to be used by this cohort. Still, it remains to be seen whether cohorts outside the youngest will adopt it.

At the semantic interpretation level, the divergence is more pronounced. For the skull, younger users assign humorous, ironic interpretations, while older cohorts either avoid the emoji entirely or assign literal or conventional meanings. For the slightly smiling and crying emoji, the divergence is simply due to usage being fully reversed: younger cohorts use them to mean passive aggression and laughter, respectively, while older cohorts use them to mean literal friendliness and crying, respectively. The sparkle emoji displays a similar trajectory but to a much lesser extent, with younger cohorts using it as a mark of sarcasm and older cohorts reading it as a sign of excitement and aesthetic readings. The wilted flower emoji's use itself gives us an insight into what the interpretation looks like- it is hardly used among cohorts other than the 18-25 group, meaning that it is not well understood or can even be taken to mean that this emoji has not penetrated wider age groups, whether or not it eventually does.

The divergences are directional, consistent, and large enough to produce the kind of systematic miscommunication described in the qualitative data. Critically, this pattern is consistent with the study's hypothesis: sarcasm-dependent emoji show sharp generational cliffs in both usage and semantic assignment. This finding strongly supports the operation of cohort-specific norms for pragmatic reanalysis, grounded in differential exposure to irony-dependent face-management strategies.

Research Objective 2: Age-of-acquisition versus current age as predictors of interpretation

As acknowledged at the design stage in Section 3.2, the survey instrument was not designed to test directly whether emoji interpretation patterns are driven by age of acquisition or by current age. The data, therefore, cannot resolve this question with the precision that confirmatory testing would require. What the data can do is examine whether the observed cohort patterns are consistent with one account.

Indirect evidence in the data is suggestive of, though not determinative for, the age-of-acquisition account. The 25-35 cohort occupies an interpretively ambiguous middle position. Its sarcasm rate (52.9%) is intermediate between the younger (78.6%) and older (around 19-22%) cohorts, and its interpretations do not align cleanly with either the pragmatic profile of the 18-25 cohort or the literal profile of the 35+ cohorts. This pattern is consistent with the prediction that this cohort, having likely adopted emoji during a period of rapid pragmatic innovation, would show a mixed interpretive repertoire reflecting partial exposure to both iconic and reanalysed meanings during their formative years. If current age were the only predictor, the 25-35 cohort would be expected to show a tidy intermediate position, which it largely does. If the age of acquisition were the sole predictor, the cohort might be expected to show a bimodal distribution between early and late adopters within the same age bracket. The current data are not sufficient to fully assess this finer-grained alternative.

To properly test the Objective 2, future research would require: (i) precise retrospective data on adoption timing and social context (age at first emoji use, peer networking at adoption, platform context); (ii) a substantially larger sample to enable stratification by both current age and adoption timing; and (iii) ideally, a prospective design in which cohorts are tracked longitudinally to determine whether interpretation patterns shift with age or remain stable at their acquisition values. The current study, designed with exploratory, rather than hypothesis-testing aims, was not adequately equipped to operationalise this distinction empirically.

The relationship between sarcasm use, face management, and pragmatic reanalysis

Beyond the two primary objectives, the data reveal a strong relationship between self-reported sarcasm use and ironic emoji interpretation that operates consistently across the sample and supports the theoretical integration of individual pragmatics and face theory. The 78.6% high sarcasm rate in the 18-25 cohort aligns with their high rates of ironic interpretation across emoji that the analysis flagged as sarcastic markers. The roughly 19% high sarcasm rate in the 35-45 cohort aligns with their predominantly literal readings. This correlation suggests that sarcasm use might be the psychological and communicative mechanism through which ironic emoji interpretation emerges and stabilises.

Following usage-based accounts of language acquisition, this finding points towards the fact that speakers who habitually deploy sarcasm as a communicative act develop interpretive defaults that assume ironic intent. They might be attuned to detecting face threats softened through indirection (Brown & Levinson) and read emoji against a baseline expectation of pragmatic reanalysis. Contrarily, speakers without sarcasm in their communicative repertoire maintain literal interpretive defaults and approach emoji meanings as transparent, iconic representations.

Contextual predictors of interpretation: Relationship, sender age, and platform

The findings were able to establish a clear hierarchy of contextual variables that shaped emoji interpretation: relationship context (reported as relevant by 78-88% across cohorts) is the strongest single predictor, followed by sender age (67-76%), then platform (around 60%), and finally message content. Notably, this hierarchy is not uniform across the cohorts. Younger users are much more sensitive to sender age as a cue to interpretation differences than older users, consistent with the theoretical prediction grounded in cohort linguistics- younger cohorts who developed their emoji competence in a heterogeneous, age-mixed online environment, have internalised a richer, more age-based set of interpretive cues than older cohorts, for whom emoji remained a novelty with limited social stratification.

The overall dominance of relationship context over sender age replicates Cui's (2022) finding in a different cultural and linguistic context, suggesting that relationship > sender age ordering may reflect a universal feature of emoji pragmatics. This finding underscores the role of face theory in predicting interpretation: interpersonal history and stakes encoded in a relationship provide the primary frame through which recipients evaluate whether an ambiguous emoji carries literal or pragmatically reanalysed meaning. Age of the sender, while significant (more so for younger recipients), operates on a secondary cue, allowing recipients to calibrate their interpretation to the plausible semantic repertoire of the sender's cohort.

5.3. Implications of the Study

The study's findings carry implications at three levels: theoretical, practical, and methodological.

Theoretical Implications

This study provides empirical support for the three-level model of generational emoji divergence as proposed in Section 2.9.

At Level 1, the finding that emoji interpretation is derived from repeated exposure to specific platforms and contexts in which these emoji are used is supportive of the usage-based learning theory. This shows that different age cohorts are exposed to different social networking/media platforms, where the use of emoji differs, and this study suggests that the interpretation of emoji may come from that difference.

At Level 2, in terms of Mannheim's 1952 concept of "fresh contact," the 18-25 age cohort again applies, as the group has no preconceived notions about emoji meanings and is, hence, free to develop new interpretations of what they see used around them. It may also be attributed to the fact that this cohort's contact came from a period of time where the general use of irony was growing on social networking platforms, which led this group to be leaning towards ironic usage and since the bridge cohort was also likely on social networking sites at the same time, they were caught between the two meaning systems of ironic usage versus literal.

Finally, at the third level, the context hierarchy finding (relationship > sender age) provides empirical grounding for the asymmetric context model hypothesis and is consistent with the face theory account of intergenerational miscommunication (Brown and Levinson, 1987; Cui, 2022).

The study also contributes to the current debate about whether generational differences in emoji interpretation are real and systematic or marginal and overstated. Some studies find relatively low variation in the interpretation of facial expression emoji, which appears to challenge the present study's findings. This might stand true, but the present study also attempts to put that notion to rest: as per the findings of the study, the data states clearly generational differences in interpretation of emoji that can be considered universally as meaning the same: the loudly crying emoji (😭) and the slightly smiling emoji (🙂) are interpreted in fundamentally opposite ways depending on which age cohort we seek the answer from. The implication is that even visually transparent emoji are subject to pragmatic reanalysis, a nd that the assumption of universal interpretation underestimates the extent to which generational variation has reshaped digital communication.

A second contribution is the identification of the bridge cohort as a productive theoretical category. Most existing work treats Millennials as a homogenous group; the present data show that the 25-35 segment is internally distinct from both younger and older cohorts in ways that deserve direct theoretical attention. The bridge cohort is offered here as an empirically grounded refinement to the cohort linguistics tradition.

Practical Implications

The study's findings are directly relevant to intergenerational communication contexts, particularly in family and workplace settings. The high rate of miscommunication reported in the sample, combined with the specific patterns of interpretive failure documented in the qualitative data, suggests that intergenerational emoji miscommunication is a real and frequent phenomenon that has practical consequences for interpersonal relationships, professional communication, and other organisational dynamics that could result in anything from momentary confusion to a larger misunderstanding that may have unexpected consequences. Two specific instances in the study's responses involving the 🙂 and 😭 emoji have been misunderstood by participants. These misreadings, if nothing else, produce some amount of anxiety (though younger users may, under scrutiny, claim that the confusion in older users is a source of amusement).

In the workplace context, the implications may be equally significant. Both the slightly smiling face and the thumbs-up emoji are very commonly used in digital communication and have undergone pragmatic reanalysis in younger cohorts, with the former serving as a marker of dismissal or hostility and the latter as a marker of passive-aggressive compliance rather than genuine agreement. Misunderstandings around these emoji can affect team dynamics and perceptions of professional rapport. While individuals can rely on contextual nuance to interpret colleagues' emoji across agr groups, the structural mismatch in cohort interpretations means misreadings will continue even with goodwill on both sides.

Several actionable responses are available. At the individual level, metacommunicative awareness (the ability to recognise that one's emoji may not be interpreted as intended) is the most direct mitigation strategy. The study's finding that 74.3% of respondents acknowledge that emoji meaning shifts based on context, while only 40% report observing this in practice. This gap between abstract and applied recognition suggests that the population has an untapped metacommunicative capacity that could be activated through deliberate attention to interpretive variation. At the organisational level, explicit discussion of digital communication norms within mixed-age teams may reduce misunderstandings, particularly around face-sensitive emoji.

It should be acknowledged that the rapid pace of emoji semantic change limits the value of any prescriptive standardisation. Younger cohorts' emerging use of the wilted flower emoji (🥀) as a marker of cringe humour has not yet propagated to older cohorts. By the time it does, innovations will likely have appeared. Generalising emoji meaning is therefore a perpetually receding target. The most realistic stance for users in mixed-age communicative settings is to be open to interpretation and to ask for clarification when an emoji's intent is unclear.

Methodological Implications

This study demonstrates that a mixed-methods questionnaire design combining Likert-scale items, contextualised interpretation tasks, and open-ended qualitative responses is a productive tool for investigating this matter at the population level. The contextualised interpretation tasks, in particular, proved among the most analytically valuable components of the instrument: by embedding emoji in naturalistic sentences and asking for specific interpretive judgements, they capture online interpretive processes in ways that general self-report questions may not. Future research in this area would benefit from building on this design, potentially supplementing it with analyses of naturally occurring message corpora or with experimental designs that systematically vary sender identity to test the contextual hierarchy hypothesis under more controlled conditions.

Contribution from a Multilingual Indian Context

A distinctive contribution of this study is its grounding in a multilingual South Asian sample. The emoji research literature has been heavily concentrated in English-speaking Western populations (Barinaga-López et al., 2022), and questions about the generalizability of generational findings across cultural contexts have remained largely open. The present study finds that the core generational patterns documented in Western contexts (sarcasm-driven divergence, the inversion of the slightly smiling face, the stratification of skull interpretation) replicate cleanly in an Indian sample whose primary languages include Telugu, Hindi, Tamil, Odia, and Marathi alongside English. This replication is itself a finding: it suggests that the generational dynamics of emoji interpretation may be relatively robust across the cultural and linguistic boundaries that separate the Anglophone West from multilingual South Asia, at least within the urban, educated populations represented here. The study, therefore, extends the empirical base of the field and points toward the value of broader international comparative research.

5.4. Limitations of the Study

This study has several limitations that define the scope and generalizability of its findings. These limitations must be addressed not as deficiencies of the study but as boundary conditions that define the domain of claims the study can legitimately support and that point toward productive directions for future research.

Sample size and Statistical Power

The total sample of n=70, along with the 45+ cohort being at n=9, limits the statistical precision of between-cohort comparisons. Percentage-point differences within the 45+ cohort must be interpreted with caution: a single additional respondent choosing a different option would shift the group's generalisations by 11%. The study's analytical strategy (descriptive cross-tabulation and qualitative pattern analysis rather than inferential testing) is appropriate given this constraint. Still, it also means that the findings cannot be reported as statistically significant in the conventional hypothesis-testing sense. Future research should aim for a minimum cell size of n=30 per cohort to support chi-square and ANOVA testing of the patterns identified here.

Sampling Bias and Population Representativeness

The convenience sample's skew towards younger, more highly educated respondents limits generalizability in three specific ways. First, because the sample is predominantly educated, the study cannot speak for how emoji divergence operates among less digitally literate or less educated populations, who may show different patterns of interpretation or acquisition. Second, the Indian cultural context introduces variables, including language, culture, regional media references, and family communication norms, that may differ from the Western individualist context. As discussed in Section 5.3, the broad replication of Western generational patterns in this sample is a finding in itself. Still, the comparison with Western data should not be over-interpreted, given the very different cultural backdrop. Third, because the 18-25 cohort is overrepresented (40% of the sample), the study may slightly overstate the prevalence of ironic emoji interpretation in the broader population.

Cohort Versus Age Effects

As noted in Section 3.6.5, the cross-sectional design cannot distinguish between cohort effects (different interpretive frameworks because users encountered emoji at different historical moments) and age effects (different interpretive frameworks because of the social contexts and communicative demands associated with each life stage). The bridge cohort analysis tentatively favours a cohort interpretation, with the 25-35 group's transitional pattern suggesting that social context, rather than maturation, is driving the semantic shift, but a longitudinal study tracking individuals over time would be required to test this interpretation directly.

Self-report Validity

The questionnaire relies entirely on self-reported emoji behaviour and interpretation. Self-report can diverge from actual behaviour due to social desirability bias, metacognitive limitations, and demand characteristics, as discussed in Section 3.6.5. The contextualised interpretation tasks in Block 3 partially mitigate this concern by capturing in-the-moment judgements rather than retrospective self-assessment, but they remain a laboratory-style measure. A complementary study using naturally occurring WhatsApp or Instagram message corpora, in which emoji use and interpretation can be observed directly instead of relying on self-report, would significantly strengthen the evidential base for the study's claims.

Emoji Selection

The five-target emoji were selected to represent a range of semantic types, from the pragmatically reanalysed skull 💀to the conventionalised 🙂. While this selection yields theoretically rich findings, it is not a representative sample of the full emoji lexicon, which comprises approximately 3,600 characters spanning a wide range of semantic domains. The patterns that were observed in this study through the five selected emoji may not generalize to emoji of the objects, food, activity, or less commonly used ones that have not undergone this pragmatic reanalysis that was the base of the paper.

In culminating these points, the limitations suggest that the findings of the study should be understood as an empirically grounded contribution to a developing conversation about generational variation in emoji interpretation, rather than as definitive conclusions. The study's primary contribution is conceptual and theoretical: the three-level framework, the bridge cohort hypothesis, and the context hierarchy finding provide a more precise and empirically grounded account of generational emoji divergence than has so far been available, and they point toward a productive research agenda that future studies can build upon.

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Appendix

Section 1: Consent

1. By continuing to fill this form, you give your consent for the information to be used in this study.

Section 2: Demographic Information

General information about the participant. Used to understand background that may have contributed to the participant's personal understanding of the emoji.

1. Age
   18–25 | 25–35 | 35–45 | 45+
2. Gender
   Male | Female | Non-binary | Other | Prefer not to say
3. First Language
   (short answer text space)
4. Language Primarily Used in Social Media
   English | Hindi | Telugu | Tamil | Other (space to specify)
5. Highest Level of Education
   Grade 10 | Grade 12 | Undergraduate | Postgraduate | Other (space to specify)
6. Do You Recall When You Began Using Emoji in Digital Communication?
   Yes | No | Maybe
7. If yes, detail when: (short answer text space)

Section 3: Digital and Social Networking Usage

Questions to understand whether there is difference between the emoji used on different platforms. To understand whether the same emoji can be used in different platforms with the same understood meaning by both receiver and sender.

1. Which Platforms Do You Regularly Use? (Select all that apply)
   WhatsApp | Instagram | X (Formerly Twitter) | Snapchat | Reddit | Facebook | Others (space to specify)
2. What Platforms Do You Spend the MOST Time On?
   WhatsApp | Instagram | X (Formerly Twitter) | Snapchat | Reddit | Facebook | Others (space to specify)
3. On Average, How Many Hours Per Day Do You Spend on Social Media?
   <1 | 1–2 | 3–4 | 5+
4. How Often Do You Use Emoji in Digital Communication?
   Never | Rarely | Sometimes | Often | Very Often
5. I feel emoji help clarify the tone in text-based communication.
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree
6. I sometimes use emoji instead of words

Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree

7. Emoji meaning changes depending on platform
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree

Section 4: General Emoji Attitudes

How the participants feel about emoji as a whole without diving deeper into specificity.

1. Emoji have stable, fixed meanings.
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree
2. Emoji meaning depends heavily on context.
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree
3. I Have Misunderstood Emoji Before
   Yes | No | Maybe
4. Emoji Meanings Have Changed Over Time Since I Started Using Them.
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree
5. Which Feeling/Emotion Do You Use Emoji to Express Most?
   Happiness | Sarcasm | Laughter | Positivity | Sadness | Danger | Negativity

Section 5: Emoji Usage Frequency

Questions to understand how often the participant uses emoji and what meaning the emoji themselves convey to the participant.

💀 Skull Emoji

1. How often do you use [💀]?
   Never | 1 | 2 | 3 | 4 | 5 (Very Often)
2. When you send [💀], what do you usually mean?
   Danger | Death | Sarcasm | Humour | Other (space to specify)

😭 Crying Loudly Emoji

1. How often do you use [😭]?
   Never | 1 | 2 | 3 | 4 | 5 (Very Often)
2. When you send [😭], what do you usually mean?
   Crying | Sadness | Sarcasm | Humour | Other (space to specify)

✨ Sparkles Emoji

1. How often do you use [✨]?
   Never | 1 | 2 | 3 | 4 | 5 (Very Often)
2. When you send [✨], what do you usually mean?
   Shiny | Sparkly | Magic | Ironic emphasis | Sarcasm | Other (space to specify)

🙂 Slightly Smiling Face Emoji

1. How often do you use [🙂]?
   Never | 1 | 2 | 3 | 4 | 5 (Very Often)
2. When you send [🙂], what do you usually mean?

Friendliness | Politeness | Passive-Aggression | Irritation | Other (space to specify)

🥀 Wilted Flower Emoji

1. How often do you use [🥀]?
   Never | 1 | 2 | 3 | 4 | 5 (Very Often)
2. When you send [🥀], what do you usually mean?
   Death | Sadness | Ironic humour | Dead flower (literal) | Other (space to specify)

Section 6: Contextual Interpretation Task

Questions to understand whether emoji meanings change based on the context. You will receive example sentences with specific emoji used in them and then you will be asked the meaning of the emoji within the context of the sentence.

1. "The presentation was amazing 💀" What does 💀 mean here?
   The presentation was deadly boring | The presentation was hilariously bad | The presentation killed it (was great) | Not sure | Other (space to specify)
2. "Stop 😭" What does 😭 mean here?
   Stop doing something because sender is sad | There is no way (negative) | There is no way (positive) | There is no way (sarcastic/ironic) | Other (space to specify)
3. "Just finished my taxes ✨" What does ✨ mean here?
   Genuine excitement | Ironic/sarcastic about a boring task | Magical feeling | Not sure | Other (space to specify)
4. "Thanks for sending the report 🙂" What does 🙂 mean here?
   Friendly and appreciative | Passive-aggressive or annoyed | Neutral/polite | Not sure | Other (space to specify)
5. "He thinks he's cool like that 🥀" What does 🥀 mean here?
   Friendly and appreciative | Passive-aggressive or annoyed | Neutral | Not sure | Other (space to specify)
6. In These Examples, Did the Emoji Meaning Change Compared to When Used Alone?
   Yes | No | Sometimes

   If yes, briefly explain why: (long answer space)

Section 7: How Your Relationship with Sender of Texts Affects Your Interpretation

1. Does the Sender's Age Affect How You Interpret Emoji?
   Yes | No | Sometimes
2. Does the Sender's Relationship to You Affect How You Interpret Emoji?
   Yes | No | Sometimes

Section 8: Intentionality and Pragmatic Strategy

1. When Choosing Emoji, I Consider:
   My relationship with the receiver | The topic of conversation | The platform | How it might be interpreted | Habit/Automatic use | Other (space to specify)

2. I use emoji sarcastically
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree
3. I use emoji to soften criticism
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree
4. I use emoji to intensify emotion
   Strongly Disagree | 1 | 2 | 3 | 4 | 5 | Strongly Agree

Section 9: Open-ended Reflection

1. Have you ever noticed older/younger people using emoji differently than you? If yes, can you give an example?
   (Long answer text space)